

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283465

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Barnett; Christopher Todd et al.

FLUID END ASSEMBLY

Abstract

A fluid end assembly having a plurality of fluid end sections positioned in a side-by-side relationship. Each fluid end section has a housing made of multiple-piece construction. One or more pieces of the housing are configured to have a plurality of stay rods attached thereto. The stay rods interconnect the fluid end assembly and a power end assembly.

Inventors: Barnett; Christopher Todd (Stratford, OK), Thomas; Micheal Cole (Azle, TX), Son; Nicholas (Davis, OK), Keith; John (Ardmore, OK), Foster; Kelcy Jake (Sulphur, OK)

Applicant: Kerr Machine Co. (Sulphur, OK)

Family ID: 1000008626565

Appl. No.: 19/219060

Filed: May 27, 2025

Related U.S. Application Data

parent US continuation 18933692 20241031 parent-grant-document US 12345254 child US 19219060

parent US continuation 16951741 20201118 parent-grant-document US 11162479 child US 17515707

parent US continuation-in-part 18490864 20231020 parent-grant-document US 12135024 child US 18933692

parent US continuation-in-part 17884712 20220810 parent-grant-document US 12188458 child US 18490864

parent US continuation-in-part 17550552 20211214 parent-grant-document US 11644018 child US 17884712

parent US continuation-in-part 17515707 20211101 parent-grant-document US 11359615 child US 17550552

parent US continuation-in-part 18488435 20231017 parent-grant-document US 12000257 child US

16951741
us-provisional-application US 62936789 20191118
us-provisional-application US 62940513 20191126
us-provisional-application US 62953763 20191226
us-provisional-application US 62957489 20200106
us-provisional-application US 62959570 20200110
us-provisional-application US 62960194 20200113
us-provisional-application US 62960366 20200113
us-provisional-application US 62968634 20200131
us-provisional-application US 62990817 20200317
us-provisional-application US 63008036 20200410
us-provisional-application US 63018021 20200430
us-provisional-application US 63019789 20200504
us-provisional-application US 63027584 20200520
us-provisional-application US 63033244 20200602
us-provisional-application US 63040086 20200617
us-provisional-application US 63046826 20200701
us-provisional-application US 63053797 20200720
us-provisional-application US 63076587 20200910
us-provisional-application US 63089882 20201009
us-provisional-application US 63125459 20201215
us-provisional-application US 63148065 20210210
us-provisional-application US 63150340 20210217
us-provisional-application US 63155835 20210303
us-provisional-application US 63168364 20210331
us-provisional-application US 63233241 20210814
us-provisional-application US 63235251 20210820
us-provisional-application US 63240889 20210904
us-provisional-application US 63246099 20210920
us-provisional-application US 63283487 20211128
us-provisional-application US 63301524 20220121
us-provisional-application US 63304070 20220128
us-provisional-application US 63309168 20220211
us-provisional-application US 63310269 20220215
us-provisional-application US 63312541 20220222
us-provisional-application US 63416644 20221017
us-provisional-application US 63422637 20221104
us-provisional-application US 63422637 20221104
us-provisional-application US 63506222 20230605
us-provisional-application US 63506222 20230605
us-provisional-application US 63508577 20230616
us-provisional-application US 63508577 20230616

Publication Classification

Int. Cl.: **F04B53/22** (20060101); **E21B43/26** (20060101); **F04B1/0421** (20200101); **F04B1/0452** (20200101); **F04B1/0538** (20200101); **F04B7/00** (20060101); **F04B7/02** (20060101); **F04B15/02** (20060101); **F04B17/00** (20060101); **F04B19/04** (20060101); **F04B19/22** (20060101); **F04B53/00** (20060101); **F04B53/10** (20060101); **F04B53/14** (20060101);

U.S. Cl.:

CPC **F04B53/22** (20130101); **F04B1/0421** (20130101); **F04B15/02** (20130101); **F04B53/007** (20130101); **F04B53/1022** (20130101); **F04B53/1032** (20130101); **F04B53/14** (20130101); **F04B53/164** (20130101); **F16J15/26** (20130101); **F16K1/42** (20130101); **F16K11/105** (20130101); E21B43/2607 (20200501); F04B1/0452 (20130101); F04B1/0538 (20130101); F04B7/0003 (20130101); F04B7/0084 (20130101); F04B7/0088 (20130101); F04B7/0208 (20130101); F04B7/0266 (20130101); F04B17/00 (20130101); F04B19/04 (20130101); F04B19/22 (20130101); F04B53/006 (20130101); F04B53/18 (20130101); F16L55/11 (20130101)

Background/Summary

[0001] This application is a Continuation-in-Part of U.S. patent application Ser. No. 17/550,552, now U.S. Pat. No. 11,644,018, authored by Thomas et al., and filed on Dec. 14, 2021, the entire contents of which are incorporated herein by reference (hereinafter “the ‘552 application”). This application is also a Continuation-in-Part of U.S. patent application Ser. No. 18/488,435, now U.S. Pat. No. 12,000,257, authored by Foster et al., and filed on Oct. 17, 2023, the entire contents of which are incorporated herein by reference. [0002] This application also claims the benefit of the following U.S. provisional patent applications: Ser. No. 63/233,241, authored by Foster et al, and filed on Aug. 14, 2021; Ser. No. 63/235,251, authored by Foster et al., and filed on Aug. 20, 2021; Ser. No. 63/240,889, authored by Foster et al., and filed on Sep. 4, 2021; Ser. No. 63/246,099, authored by Foster et al., and filed on Sep. 20, 2021; Ser. No. 63/301,524, authored by Foster et al., and filed on Jan. 21, 2022; Ser. No. 63/304,070, authored by Foster et al., and filed on Jan. 28, 2022; Ser. No. 63/310,269, authored by Foster et al., and filed on Feb. 15, 2022; Ser. No. 63/312,541, authored by Foster et al., and filed on Feb. 22, 2022; Ser. No. 63/416,644, authored by Foster et al., and filed on Oct. 17, 2022; Ser. No. 63/422,637, authored by Foster et al., and filed on Nov. 4, 2022; Ser. No. 63/506,222, authored by Foster et al., and filed on Jun. 5, 2023; and Ser. No. 63/508,577, authored by Foster et al., and filed on Jun. 16, 2023. The entire contents of all of the above listed provisional patent applications are incorporated herein by reference.

BACKGROUND

[0003] Various industrial applications may require the delivery of high volumes of highly pressurized fluids. For example, hydraulic fracturing (commonly referred to as “fracking”) is a well stimulation technique used in oil and gas production, in which highly pressurized fluid is injected into a cased wellbore. As shown for example in FIG. 1, the pressured fluid flows through perforations **10** in a casing **12** and creates fractures **14** in deep rock formations **16**. Pressurized fluid is delivered to the casing **12** through a wellhead **18** supported on the ground surface **20**. Sand or other small particles (commonly referred to as “proppants”) are normally delivered with the fluid into the rock formations **16**. The proppants help hold the fractures **14** open after the fluid is withdrawn. The resulting fractures **14** facilitate the extraction of oil, gas, brine, or other fluid trapped within the rock formations **16**.

[0004] Fluid ends are devices used in conjunction with a power source to pressurize the fluid used during hydraulic fracturing operations. A single fracking operation may require the use of two or more fluid ends at one time. For example, six fluid ends **22** are shown operating at a wellsite **24** in FIG. 2. Each of the fluid ends **22** is attached to a power end **26** in a one-to-one relationship. The power end **26** serves as an engine or motor for the fluid end **22**. Together, the fluid end **22** and

power end **26** function as a high pressure pump.

[0005] Continuing with FIG. **2**, a single fluid end **22** and its corresponding power end **26** are typically positioned on a truck bed **28** at the wellsite **24** so that they may be easily moved, as needed. The fluid and proppant mixture to be pressurized is normally held in large tanks **30** at the wellsite **24**. An intake piping system **32** delivers the fluid and proppant mixture from the tanks **30** to each fluid end **22**. A discharge piping system **33** transfers the pressurized fluid from each fluid end **22** to the wellhead **18**, where it is delivered into the casing **12** shown in FIG. **1**.

[0006] Fluid ends operate under notoriously extreme conditions, enduring the same pressures, vibrations, and abrasives that are needed to fracture the deep rock formations **16**, shown in FIG. **1**. Fluid ends may operate at pressures of 5,000-15,000 pounds per square inch (psi) or greater. Fluid used in hydraulic fracturing operations is typically pumped through the fluid end at a pressure of at least 8,000 psi, and more typically between 10,000 and 15,000 psi. However, the pressure may reach up to 22,500 psi. The power end used with the fluid end typically has a power output of at least 2,250 horsepower during hydraulic fracturing operations. A single fluid end typically produces a fluid volume of about 400 gallons, or 10 barrels, per minute during a fracking operation. A single fluid end may operate in flow ranges from 170 to 630 gallons per minute, or approximately 4 to 15 barrels per minute. When a plurality of fluid ends are used together, the fluid ends collectively may deliver as much as 4,200 gallons per minute or 100 barrels per minute to the wellbore.

[0007] In contrast, mud pumps known in the art typically operate at a pressure of less than 8,000 psi. Mud pumps are used to deliver drilling mud to a rotating drill bit within the wellbore during drilling operations. Thus, the drilling mud does not need to have as high of fluid pressure as fracking fluid. A fluid end does not pump drilling mud. A power end used with mud pumps typically has a power output of less than 2,250 horsepower. Mud pumps generally produce a fluid volume of about 150-600 gallons per minute, depending on the size of pump used.

[0008] In further contrast, a fluid jetting pump known in the art typically operates at pressures of 30,000-90,000 psi. Jet pumps are used to deliver a highly concentrated stream of fluid to a desired area. Jet pumps typically deliver fluid through a wand. Fluid ends do not deliver fluid through a wand. Unlike fluid ends, jet pumps are not used in concert with a plurality of other jet pumps. Rather, only a single jet pump is used to pressurize fluid. A power end used with a jet pump typically has a power output of about 1,000 horsepower. Jet pumps generally produce a fluid volume of about 10 gallons per minute.

[0009] High operational pressures may cause a fluid end to expand or crack. Such a structural failure may lead to fluid leakage, which leaves the fluid end unable to produce and maintain adequate fluid pressures. Moreover, if proppants are included in the pressurized fluid, those proppants may cause erosion at weak points within the fluid end, resulting in additional failures.

[0010] It is not uncommon for conventional fluid ends to experience failure after only several hundred operating hours. Yet, a single fracking operation may require as many as fifty (50) hours of fluid end operation. Thus, a traditional fluid end may require replacement after use on as few as two fracking jobs. There is a need in the industry for a fluid end configured to avoid or significantly delay the structures or conditions that cause wear or failures within a fluid end.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

First Embodiment

[0011] FIG. **1** is an illustration of the underground environment of a hydraulic fracturing operation.

[0012] FIG. **2** illustrates above-ground equipment used in a hydraulic fracturing operation.

[0013] FIG. **3** is a front perspective view of one embodiment of a high pressure pump disclosed herein.

[0014] FIG. 4 is a front perspective view of the fluid end assembly shown in FIG. 3 attached to a plurality of stay rods.

[0015] FIG. 5 is a rear perspective view of the fluid end assembly and stay rods shown in FIG. 4.

[0016] FIG. 6 is a front elevational view of the fluid end assembly and stay rods shown in FIG. 4.

[0017] FIG. 7 is a front perspective view of the fluid end assembly shown in FIGS. 3 and 4.

[0018] FIG. 8 is a rear perspective view of the fluid end assembly shown in FIG. 7.

[0019] FIG. 9 is a cross-sectional view of one of the fluid end sections making up the fluid end assembly shown in FIG. 6, taken along lines A-A.

[0020] FIG. 10 is a front perspective view of one of the housings used with one of the fluid end sections shown in FIG. 7.

[0021] FIG. 11 is a rear perspective view of the housing shown in FIG. 10.

[0022] FIG. 12 is a front elevational view of the housing shown in FIG. 10.

[0023] FIG. 13 is a rear elevational view of the housing shown in FIG. 10.

[0024] FIG. 14 is a cross-sectional view of the housing shown in FIG. 13, taken along line B-B.

[0025] FIG. 15 is a cross-sectional view of the housing shown in FIG. 13, taken along line C-C.

[0026] FIG. 16 is a cross-sectional view of the housing shown in FIG. 13, taken along line D-D.

[0027] FIG. 17 is a front perspective view of the first section of the housing shown in FIG. 10.

[0028] FIG. 18 is a rear perspective view of the first section shown in FIG. 17.

[0029] FIG. 19 is a front perspective view of the first section shown in FIG. 17, but the first section has a plurality of stay rods attached thereto.

[0030] FIG. 20 is a rear perspective view of the first section and stay rods shown in FIG. 19.

[0031] FIG. 21 is a cross-sectional view of the first section and stay rods shown in FIG. 19, taken along line E-E.

[0032] FIG. 22 is an enlarged view of area F shown in FIG. 21.

[0033] FIG. 23 is a front perspective view of a second section of the housing shown in FIG. 10.

[0034] FIG. 24 is a rear perspective view of the second section shown in FIG. 23.

[0035] FIG. 25 is a front perspective view of the third section of the housing shown in FIG. 10.

[0036] FIG. 26 is a rear perspective view of the third section shown in FIG. 25.

[0037] FIG. 27 is a front perspective and exploded view of the housing shown in FIG. 10.

[0038] FIG. 28 is a rear perspective and exploded view of the housing shown in FIG. 10.

[0039] FIG. 29 is a cross-sectional view of the fluid end assembly shown in FIG. 6, taken along line G-G.

[0040] FIG. 30 is a rear perspective view of the retention plate shown in FIGS. 9 and 29.

[0041] FIG. 31 is a front perspective view of the retention plate shown in FIG. 30.

[0042] FIG. 32 is a rear perspective view of the retention plate shown in FIG. 30 attached to the housing shown in FIG. 10.

[0043] FIG. 33 is a front perspective view of the stuffing box shown in FIGS. 9 and 29.

[0044] FIG. 34 is a rear perspective view of the stuffing box shown in FIG. 33.

[0045] FIG. 35 is a front perspective view of the rear retainer shown in FIGS. 9 and 29.

[0046] FIG. 36 is a rear perspective view of the rear retainer shown in FIG. 35.

[0047] FIG. 37 is a front perspective view of the packing nut shown in FIGS. 9 and 29.

[0048] FIG. 38 is a rear perspective view of the packing nut shown in FIG. 37.

[0049] FIG. 39 is an enlarged cross-sectional view of the components attached to a rear surface of the housing in FIG. 9.

[0050] FIG. 40 is the cross-sectional view of the fluid end assembly shown in FIG. 9. The plunger is retracted, and the discharge valve is shown in a closed position.

[0051] FIG. 41 is the cross-sectional view of the fluid end assembly shown in FIG. 40, but the plunger is extended within the fluid end section and the suction valve is in a closed position.

[0052] FIG. 42 is an enlarged view of area I shown in FIG. 41.

[0053] FIG. 43 is a front perspective view of the fluid routing plug shown in FIGS. 40 and 41.

[0054] FIG. **44** is a top plan view of the fluid routing plug shown in FIG. **43**.
[0055] FIG. **45** is a front elevational view of the fluid routing plug shown in FIG. **43**.
[0056] FIG. **46** is a rear perspective view of the fluid routing plug shown in FIG. **43**.
[0057] FIG. **47** is a rear elevational view of the fluid routing plug shown in FIG. **43**.
[0058] FIG. **48** is a cross-sectional view of the fluid routing plug shown in FIG. **47**, taken along line J-J.
[0059] FIG. **49** is a top perspective view of the fluid routing plug shown in FIG. **43**.
[0060] FIG. **50** is an enlarged view of area K shown in FIG. **49**.
[0061] FIG. **51** is a perspective cross-sectional view of the fluid routing plug shown in FIG. **44**, taken along line L-L.
[0062] FIG. **52** is a perspective cross-sectional view of the fluid routing plug shown in FIG. **44**, taken along line M-M.
[0063] FIG. **53** is a perspective cross-sectional view of the fluid routing plug shown in FIG. **44**, taken along line N-N.
[0064] FIG. **54** is a rear elevational view of the fluid routing plug shown in FIG. **43**, but one of the suction passages and one of the discharge passages are shown in phantom.
[0065] FIG. **55** is a cross-sectional view of the fluid routing plug shown in FIG. **44**, taken along line M-M.
[0066] FIG. **56** is a front elevational and conical-sectional view of the fluid routing plug shown in FIG. **43**. The conical-section is taken from line O in FIGS. **43** and **45** to line P in FIGS. **46** and **47**.
[0067] FIG. **57** is a front perspective and conical-sectional view of the fluid routing plug shown in FIG. **56**.
[0068] FIG. **58** is a rear perspective and conical-sectional view of the fluid routing plug shown in FIG. **56**.
[0069] FIG. **59** is a side elevational and conical-sectional view of the fluid routing plug shown in FIG. **56**.
[0070] FIG. **60** is a side elevational view of the hardened insert shown in FIGS. **40** and **41**.
[0071] FIG. **61** is a front perspective view of the hardened insert shown in FIG. **60**.
[0072] FIG. **62** is a front elevational view of the hardened insert shown in FIG. **60**.
[0073] FIG. **63** is a cross-sectional view of the hardened insert shown in FIG. **62**, taken along line Q-Q.
[0074] FIG. **64** is a front perspective view of the discharge valve shown in FIGS. **40** and **41**.
[0075] FIG. **65** is a rear perspective view of the discharge valve shown in FIG. **64**.
[0076] FIG. **66** is a side elevational view of the discharge valve shown in FIG. **64**.
[0077] FIG. **67** is a cross-sectional view of the discharge valve shown in FIG. **66**, taken along line R-R.
[0078] FIG. **68** is a front perspective view of the suction valve guide shown in FIGS. **40** and **41**.
[0079] FIG. **69** is a rear perspective view of the suction valve guide shown in FIG. **68**.
[0080] FIG. **70** is a front elevational view of the suction valve guide shown in FIG. **68**.
[0081] FIG. **71** is a cross-sectional view of the suction valve guide shown in FIG. **70**, taken along line S-S.
[0082] FIG. **72** is a front perspective view of the discharge plug shown in FIGS. **40** and **41**.
[0083] FIG. **73** is a rear perspective view of the discharge plug shown in FIG. **72**.
[0084] FIG. **74** is a front elevational view of the discharge plug shown in FIG. **72**.
[0085] FIG. **75** is a cross-sectional view of the discharge plug shown in FIG. **74**, taken along line T-T.
[0086] FIG. **76** is a front perspective view of the front retainer shown in FIGS. **40** and **41**.
[0087] FIG. **77** is a rear perspective view of the front retainer shown in FIG. **76**.
[0088] FIG. **78** is an enlarged view of area U shown in FIG. **41**.
[0089] FIG. **79** is a front perspective and exploded view of the fluid end section shown in FIGS. **9**,

29, 40, and 41.

[0090] FIG. 80 is a rear perspective and exploded view of the fluid end section shown in FIGS. 9, 29, 40, and 41.

Alternative Embodiments

[0091] FIG. 81 is a front perspective view of another embodiment of a fluid end section.

[0092] FIG. 82 is a rear perspective view of the fluid end section shown in FIG. 81.

[0093] FIG. 83 is a front perspective view of the fluid end section shown in FIG. 81 having a plurality of stay rods attached thereto.

[0094] FIG. 84 is a rear perspective view of the fluid end section and stay rods shown in FIG. 83.

[0095] FIG. 85 is a cross-sectional view of the fluid end section and stay rods shown in FIGS. 83 and 84, taken along line V-V.

[0096] FIG. 86 is a cross-sectional view of the fluid end section shown in FIGS. 81 and 82, taken along line W-W.

[0097] FIG. 87 is a rear perspective and partially exploded view of the fluid end section shown in FIG. 81.

[0098] FIG. 88 is a cross-sectional view of the fluid end section shown in FIGS. 81 and 82, taken along line X-X.

[0099] FIG. 89 is a top plan view of another embodiment of a fluid routing plug. The fluid routing plug is shown installed within the fluid end section in FIGS. 86 and 88.

[0100] FIG. 90 is a front perspective view of the fluid routing plug shown in FIG. 89.

[0101] FIG. 91 is a front elevational view of the fluid routing plug shown in FIG. 89.

[0102] FIG. 92 is a cross-sectional view of the fluid routing plug shown in FIG. 91, taken along line Y-Y.

[0103] FIG. 93 is an enlarged view of area Z shown in FIG. 88.

[0104] FIG. 94 is an enlarged view of area AA shown in FIG. 92.

[0105] FIG. 95 is a front perspective view of another embodiment of a fluid end section.

[0106] FIG. 96 is a rear perspective view of the fluid end section shown in FIG. 95.

[0107] FIG. 97 is a front perspective view of the integral section of the housing shown in FIG. 95.

[0108] FIG. 98 is a rear perspective view of the integral section shown in FIG. 97.

[0109] FIG. 99 is a front perspective view of the fluid end section shown in FIG. 95 having a plurality of stay rods attached thereto.

[0110] FIG. 100 is a rear perspective view of the fluid end section and stay rods shown in FIG. 99.

[0111] FIG. 101 is a cross-sectional view of the fluid end section and stay rods shown in FIGS. 99 and 100, taken along lines AB-AB.

[0112] FIG. 102 is a cross-sectional view of the fluid end section shown in FIGS. 95 and 96, taken along lines AC-AC.

[0113] FIG. 103 is an enlarged view of area AD shown in FIG. 122.

[0114] FIG. 104 is a front perspective view of another embodiment of a fluid routing plug. The fluid routing plug is shown installed within the fluid end section in FIG. 103.

[0115] FIG. 105 is a front elevational view of the fluid routing plug shown in FIG. 104.

[0116] FIG. 106 is a side elevational view of the fluid routing plug shown in FIG. 104.

[0117] FIG. 107 is a rear perspective view of the fluid routing plug shown in FIG. 104.

[0118] FIG. 108 is a cross-sectional view of the fluid routing plug shown in FIG. 105, taken along line AE-AE.

[0119] FIG. 109 is a front perspective view of another embodiment of a fluid end section.

[0120] FIG. 110 is a rear perspective view of the fluid end section shown in FIG. 109.

[0121] FIG. 111 is a cross-sectional view of the fluid end section shown in FIGS. 109 and 110, taken along line AF-AF.

[0122] FIG. 112 is a cross-sectional view of the fluid end section shown in FIGS. 109 and 110, taken along line AG-AG.

[0123] FIG. **113** is an enlarged view of area AH shown in FIG. **111**.

[0124] FIG. **114** is a front perspective view of the support element installed within the packing seal shown in FIG. **113**.

[0125] FIG. **115** is a cross-sectional view of the support element and packing seal shown in FIG. **114**, taken along line AI-AI.

[0126] FIG. **116** is an enlarged view of area AJ shown in FIG. **115**.

[0127] FIG. **117** is a front perspective view of another embodiment of a fluid end section.

[0128] FIG. **118** is a rear perspective view of the fluid end section shown in FIG. **117**.

[0129] FIG. **119** is a front perspective and partially exploded view of the fluid end section shown in FIG. **117**.

[0130] FIG. **120** is a cross-sectional view of the fluid end assembly shown in FIG. **124**, taken along line AK-AK.

[0131] FIG. **121** is an enlarged view of area AL shown in FIG. **120**.

[0132] FIG. **122** is a cross-sectional view of the fluid end assembly shown in FIG. **124**, taken along line AM-AM.

[0133] FIG. **123** is an enlarged view of area AN shown in FIG. **122**.

[0134] FIG. **124** is a front perspective view of another embodiment of a high-pressure pump. The fluid end assembly uses the fluid end section shown in FIG. **117**.

[0135] FIG. **125** is a front perspective view of another embodiment of a fluid end section.

[0136] FIG. **126** is a rear perspective view of the fluid end section shown in FIG. **125**.

[0137] FIG. **127** is a cross-sectional view of the fluid end section shown in FIG. **126**, taken along line AO-AO.

[0138] FIG. **128** is a front perspective view of the third section of the housing used with the fluid end section shown in FIG. **125**.

[0139] FIG. **129** is a rear perspective view of the third section of the housing shown in FIG. **128**.

[0140] FIG. **130** is a cross-sectional view of the third section of the housing shown in FIG. **128**, taken along line AP-AP.

[0141] FIG. **131** is a cross-sectional view of the third section of the housing shown in FIG. **129**, taken along line AQ-AQ.

[0142] FIG. **132** is a rear perspective view of a blind nut used with the fluid end section shown in FIG. **125**.

[0143] FIG. **133** is a front perspective view of the blind nut shown in FIG. **132**.

[0144] FIG. **134** is a rear elevational view of the blind nut shown in FIG. **132**.

[0145] FIG. **135** is a cross-sectional view of the blind nut shown in FIG. **134**, taken along line AR-AR.

[0146] FIG. **136** is a front perspective view of a reaction washer used with the fluid end section shown in FIG. **125**.

[0147] FIG. **137** is a front elevational view of the reaction washer shown in FIG. **136**.

[0148] FIG. **138** is a cross-sectional view of the fluid end section shown in FIG. **125**, taken long line AS-AS.

[0149] FIG. **139** is an enlarged view of area AT, shown in FIG. **127**.

[0150] FIG. **140** is an enlarged view of area AU, shown in FIG. **127**.

[0151] FIG. **141** is an enlarged view of area AV, shown in FIG. **127**.

[0152] FIG. **142** is a front perspective and exploded view of the fluid end section shown in FIG. **125**.

[0153] FIG. **143** is a front perspective view of another embodiment of a fluid end section.

[0154] FIG. **144** is a rear perspective view of the fluid end section shown in FIG. **143**.

[0155] FIG. **145** is a cross-sectional view of the fluid end section shown in FIG. **143**, taken along line AW-AW, but the plunger is shown retracted.

[0156] FIG. **146** is a cross-sectional view of the fluid end section shown in FIG. **144**, taken long

line AX-AX.

[0157] FIG. **147** is a cross-sectional view of the fluid end section shown in FIG. **144**, taken along line AY-AY.

[0158] FIG. **148** is a front perspective view of the housing used with the fluid end section shown in FIG. **143**.

[0159] FIG. **149** is a rear perspective view of the housing shown in FIG. **148**.

[0160] FIG. **150** is a cross-sectional view of the housing shown in FIG. **148**, taken along line AZ-AZ.

[0161] FIG. **151** is a cross-sectional view of the housing shown in FIG. **149**, taken along line BA-BA.

[0162] FIG. **152** is a front perspective view of the first section of the housing shown in FIG. **148**.

[0163] FIG. **153** is a rear perspective view of the first section of the housing shown in FIG. **152**.

[0164] FIG. **154** is a front perspective view of the first section of the housing shown in FIG. **152** with a plurality of stay rods attached thereto.

[0165] FIG. **155** is a rear perspective view of the first section of the housing and stay rods shown in FIG. **154**.

[0166] FIG. **156** is a front perspective view of the second section of the housing shown in FIG. **148**.

[0167] FIG. **157** is a rear perspective view of the second section of the housing shown in FIG. **156**.

[0168] FIG. **158** is a front perspective view of the front spacer sleeve used with the housing shown in FIG. **148**.

[0169] FIG. **159** is a rear perspective view of the front spacer sleeve shown in FIG. **158**.

[0170] FIG. **160** is a front perspective and exploded view of the housing shown in FIG. **148**.

[0171] FIG. **161** is a rear perspective and exploded view of the housing shown in FIG. **148**.

[0172] FIG. **162** is a front perspective view of the rear spacer sleeve used with the fluid end section shown in FIGS. **145-147**.

[0173] FIG. **163** is a rear perspective view of the rear spacer sleeve shown in FIG. **162**.

[0174] FIG. **164** is an enlarged view of area BB, shown in FIG. **145**.

[0175] FIG. **165** is an enlarged view of area BC, shown in FIG. **164**.

[0176] FIG. **166** is an enlarged view of area BD, shown in FIG. **147**.

[0177] FIG. **167** is a front perspective and exploded view of the fluid end section shown in FIG. **143**.

[0178] FIG. **168** is a rear perspective and exploded view of the fluid end section shown in FIG. **167**.

[0179] FIG. **169** is a front perspective view of another embodiment of a fluid end section.

[0180] FIG. **170** is a rear perspective view of the fluid end section shown in FIG. **169**.

[0181] FIG. **171** is a cross-sectional view of the fluid end section shown in FIG. **170**, taken along line BE.

[0182] FIG. **172** is the cross-sectional view of the fluid end section shown in FIG. **171** with the inner components shown installed therein.

[0183] FIG. **173** is a front perspective and exploded view of the fluid end section shown in FIG. **169**.

[0184] FIG. **174** is a rear perspective and exploded view of the fluid end section shown in FIG. **173**.

[0185] FIG. **175** is an enlarged view of area BF, shown in FIG. **165**.

[0186] FIG. **176** is an enlarged view of area BG, shown in FIG. **48**. All elements of the view behind the section plane have been removed for clarity.

[0187] FIG. **177** is an enlarged view of area BF, shown in FIG. **165**. Another embodiment of the stress relief cutout is shown on the fluid routing plug.

[0188] FIG. **178** is an enlarged view of area BG, shown in FIG. **48**. This is the same embodiment of

stress relief cutout shown in FIG. 177. All elements of the view behind the section plane have been removed for clarity.

[0189] FIG. 179 is an enlarged view of area BF, shown in FIG. 165. Another embodiment of the stress relief cutout is shown on the fluid routing plug.

[0190] FIG. 180 is an enlarged view of area BG, shown in FIG. 48. This is the same embodiment of stress relief cutout shown in FIG. 179. All elements of the view behind the section plane have been removed for clarity.

[0191] FIG. 181 is an enlarged view of area BF, shown in FIG. 165. Another embodiment of the stress relief cutout is shown on the fluid routing plug.

[0192] FIG. 182 is an enlarged view of area BG, shown in FIG. 48. This is the same embodiment of stress relief cutout shown in FIG. 181. All elements of the view behind the section plane have been removed for clarity.

[0193] FIG. 183 is an enlarged view of area BF, shown in FIG. 165. Another embodiment of the stress relief cutout is shown on the fluid routing plug.

[0194] FIG. 184 is an enlarged view of area BG, shown in FIG. 48. This is the same embodiment of stress relief cutout shown in FIG. 183. All elements of the view behind the section plane have been removed for clarity.

[0195] FIG. 185 is an enlarged view of area BF, shown in FIG. 165. Another embodiment of stress relief cutout is shown on the hardened insert.

[0196] FIG. 186 is an enlarged view of area BH, shown in FIG. 63. This is the same embodiment of stress relief cutout shown in FIG. 185. All elements of the view behind the section plane have been removed for clarity.

DETAILED DESCRIPTION

Fluid End Assembly

[0197] Turning now to FIGS. 3-80, a high pressure pump 50 is shown in FIG. 3. The pump 50 comprises a fluid end assembly 52 joined to a power end assembly 54. The power end assembly 54 is described in more detail in U.S. patent application Ser. No. 17/884,691, authored by Keith et al., and filed on Aug. 10, 2022, the entire contents of which are incorporated herein by reference (hereinafter “the '691 application”). In alternative embodiments, the fluid end assembly 52 may be attached to other power end designs known in the art. The fluid end assembly 52 and its various embodiments are described herein.

Fluid End Section

[0198] Turning to FIGS. 4-8, the fluid end assembly 52 comprises a plurality of fluid end sections 56 positioned in a side-by-side relationship, as shown in FIGS. 6-8. Each fluid end section 56 is attached to the power end assembly 54 using a plurality of stay rods 58, as shown in FIGS. 4 and 5. Preferably, the fluid end assembly 52 comprises five fluid end sections 56 positioned adjacent one another. In alternative embodiments, the fluid end assembly 52 may comprise more or less than five fluid end sections 56. In operation, a single fluid end section 56 may be removed and replaced without removing the other fluid end sections 56 from the fluid end assembly 52.

Housing of Fluid End Section

[0199] Turning to FIGS. 10-16, each fluid end section 56 comprises a horizontally positioned housing 60 having a longitudinal axis 62 extending therethrough, as shown in FIGS. 10 and 11. The housing 60 has opposed front and rear surfaces 64 and 66 joined by an outer intermediate surface 68. A horizontal bore 70 is formed within the housing 60 and interconnects the front and rear surfaces 64 and 66, as shown in FIGS. 14 and 15. The horizontal bore 70 is sized to receive various components configured to route fluid throughout the housing 60, as shown in FIG. 9. The various components will be described in more detail later herein.

[0200] Continuing with FIGS. 10-16, the housing 60 is of multi-piece construction. The housing 60 comprises a first section 72 joined to a second section 74 and a third section 76 by a plurality of first fasteners 78, as shown in FIGS. 15 and 16. By making the housing 60 out of multiple pieces

rather than a single, integral piece, any one of the sections **72**, **74**, and **76** may be removed and replaced with a new section **72**, **74**, and **76**, without replacing the other sections. For example, if a portion of the second section **74** begins to erode or crack, the second section **74** can be replaced without having to replace the first or third sections **72** and **76**. In contrast, if the housing **60** were one single piece, the entire housing would need to be replaced, resulting in much more costly repair to the fluid end assembly **52**.

First Section of Housing

[0201] Turning to FIGS. **17-21**, the first section **72** is positioned at the front end of the housing **60** and includes the front surface **64**. During operation, fluid within the first section **72** remains at relatively the same high pressure. Thus, the first section **72** is considered the static or constant high pressure section of the housing **60**. The first section **72** is configured to be attached to a plurality of the stay rods **58**, as shown in FIGS. **19-21**. Thus, each fluid end section **56** is attached to the power end assembly **54** via the first section **72** of the housing **60**.

[0202] Continuing with FIGS. **17** and **18**, each first section **72** comprises the front surface **64** joined to a rear surface **80**. The surfaces **64** and **80** are interconnected by a portion of the outer intermediate surface **68** and a portion of the horizontal bore **70**. The outer intermediate surface **68** of the first section **72** has the shape of a rectangular prism with a plurality of notches **82** formed within the front surface **64**. A notch **82** is formed within each corner of the first section **72** such that the front surface **64** has a cross-sectional shape of a cross sign having radiused corners. The notches **82** are configured to receive a first end **84** of each stay rod **58**, as shown in FIG. **21**.

[0203] With reference to FIGS. **17-21**, a plurality of passages **86** are formed in the first section **72**. Each passage **86** interconnects the rear surface **80** and a medial surface **88** of the first section **72**. The medial surface **88** is defined by the plurality of notches **82**. Each passage **86** comprises a counterbore **87** that opens on the rear surface **80**, as shown in FIG. **16**, and is configured to receive a corresponding one of the stay rods **58**. When installed within the first section **72**, the first end **84** of each stay rod **58** projects from the medial surface **88** and into the corresponding notch **82**, as shown in FIG. **21**.

[0204] Continuing with FIGS. **19-21**, a threaded nut **90** is installed on the first end **84** of each stay rod **58** within each notch **82**. The nut **90** is a three-piece nut, also known as a torque nut, which facilitates the application of high torque required to properly fasten the fluid end section **56** to the power end assembly **54**. The nut **90** is described in more detail in the '691 application. In alternative embodiments, a traditional 12-point flange nut similar to the flange nut **230**, shown in FIGS. **27** and **28**, may be installed on the first end **84** of each stay rod **58** instead of the nut **90**.

[0205] Continuing with FIGS. **19-22**, a sleeve **94** is disposed around a portion of each stay rod **58** and extends between the rear surface **80** of the first section **72** and the power end assembly **54**, as shown in FIG. **3**. A dowel sleeve **93** is inserted into each counterbore **87** formed in each passage **86**, as shown in FIG. **22**. When installed therein, a portion of the dowel sleeve **93** projects from the rear surface **80** of the first section **72**. A counterbore **95** is formed within the hollow interior of the sleeve **94** for receiving the projecting end of the dowel sleeve **93**, as shown in FIG. **22**. The dowel sleeve **93** aligns the sleeve **94** and the passage **86** concentrically. Such alignment maintains a planar engagement between the rear surface **80** of the first section **72** and the sleeve **94**. When the nut **90** is torqued against the medial surface **88** of the first section **72**, the sleeve **94** abuts the rear surface **80** of the first section **72**, rigidly securing the first section **72** to the stay rod **58**.

[0206] Turning back to FIGS. **14-16**, and **18**, a plurality of threaded openings **96** are formed in the rear surface **80** of the first section **72**. The openings **96** surround an opening of the horizontal bore **70**, as shown in FIG. **18**. Each opening **96** is configured to receive a corresponding one of the first fasteners **78** used to secure the sections **72**, **74**, and **76** together, as shown in FIGS. **15** and **16**. A plurality of dowel openings **98** are also formed in the rear surface **80** adjacent the openings **96**, as shown in FIGS. **14** and **18**. The dowel openings **98** are configured to receive first alignment dowels **100**, as shown in FIG. **14**. The first alignment dowels **100** assist in properly aligning the first

section 72 and the second section 74 during assembly of the housing 60.

[0207] Continuing with FIG. 14, a pair of upper and lower discharge bores 102 and 104 are formed within the first section 72 and interconnect the intermediate surface 68 and the horizontal bore 70. The upper and lower discharge bores 102 and 104 shown in FIG. 14 are collinear. In alternative embodiments, the bores 102 and 104 may be offset from one another and not collinear. Each bore 102 and 104 may include a counterbore 106 that opens on the intermediate surface 68. Each counterbore 106 is sized to receive a portion of a discharge fitting adapter 504, as shown in FIG. 9. The fitting adapter 504 spans between the discharge bore 102 or 104 and a discharge fitting 108 attached to the outer intermediate surface 68 of the first section 72.

[0208] Continuing with FIG. 14, a groove 110 may be formed in the side walls of the counterbore 106 for receiving a seal 112. The seal 112 engages an outer surface of the fitting adapter 504 to prevent fluid from leaking between the first section 72 and the discharge fitting 108, as shown in FIG. 9.

[0209] With reference to FIGS. 17 and 18, a plurality of threaded openings 114 are formed in the intermediate surface 68 and surrounding the opening of the upper and lower discharge bores 102 and 104. The threaded openings 114 are configured to receive a plurality of threaded fasteners 116 configured to secure a discharge fitting 108 to the first section 72, as shown in FIG. 9.

[0210] Continuing with FIGS. 14 and 15, the walls of the horizontal bore 70 within the first section 72 and positioned between the front surface 64 and the upper and lower discharge bores 102 and 104 are sized to receive a front retainer 118 and a discharge plug 120, as shown in FIG. 9. The discharge plug 120 seals fluid from leaking from the front surface 64 of the housing 60, and the front retainer 118 secures the discharge plug 120 within the first section 72 of the housing 60.

[0211] Continuing with FIGS. 14 and 15, internal threads 122 are formed in the walls of the horizontal bore 70 of the first section 72 for mating with external threads 124, shown in FIGS. 76 and 77, formed on an outer surface of the front retainer 118. In contrast, an outer surface of the discharge plug 120 faces the smooth walls of the horizontal bore 70 of the first section 72. A small amount of clearance may exist between the plug 120 and the walls of the horizontal bore 70 of the first section 72.

[0212] Continuing with FIGS. 14 and 15, a groove 125 may be formed in such walls for receiving a seal 126 configured to engage an outer surface of the discharge plug 120, as shown in FIG. 9. The seal 126 prevents fluid from leaking around the discharge plug 120 during operation. A locating cutout 128 may further be formed in the walls that is configured to receive a locating dowel pin 130. As will be described later herein, the locating dowel pin 130 is used to properly align the discharge plug 120 within the housing 60.

[0213] Continuing with FIGS. 14 and 15, the walls of the horizontal bore 70 positioned between the upper and lower discharge bores 102 and 104 and the rear surface 80 of the first section 72 are sized to receive a portion of a fluid routing plug 132, as shown in FIG. 9. This area of the walls of the horizontal bore 70 includes a counterbore 134 that opens on the rear surface 80. The counterbore 134 is sized to receive a wear ring 136, as shown in FIG. 9. The wear ring 136 has an annular shape and is configured to engage a first seal 386 installed within an outer surface of the fluid routing plug 132, as shown in FIG. 9. In alternative embodiments, the first section 72 may not include the counterbore 134 or the wear ring 136 and instead may be sized to directly engage the first seal 386 installed within the fluid routing plug 132.

[0214] Continuing with FIG. 9, in addition to the above mentioned components, the first section 72 is also configured to house a discharge valve 138. The components discussed above and installed within the first section 72 will be described in more detail later herein.

Second Section of Housing

[0215] Turning to FIGS. 23 and 24, the second section 74 of the housing 60 is configured to be positioned between the first and third sections 72 and 76 and has a cylindrical cross-sectional shape. During operation, fluid pressure within the second section 74 remains at relatively the same

pressure. The pressure is lower than that within the first section **72**. Thus, the second section **74** may be referred to as the static or constant low pressure section of the housing **60**. The second section **74** comprises opposed front and rear surfaces **140** and **142** joined by a portion of the outer intermediate surface **68** and a portion of the horizontal bore **70**.

[0216] Continuing with FIGS. **15**, **16**, **23**, and **24**, a plurality of passages **144** are formed in the second section **74**. The passages **144** surround the horizontal bore **70** and interconnect the front and rear surfaces **140** and **142**, as shown in FIGS. **15** and **16**. Each passage **144** is configured to receive a corresponding one of the first fasteners **78** used to secure the sections **72**, **74**, and **76** of the housing **60** together.

[0217] Continuing with FIGS. **14**, **23**, and **24**, a plurality of dowel openings **146** are formed in the front surface **140** of the second section **74**, as shown in FIG. **23**. The dowel openings **146** align with the dowel openings **98** formed in the rear surface **80** of the first section **72** and are configured to receive a portion of the first alignment dowels **100**, as shown in FIG. **14**. Likewise, a plurality of dowel openings **148** are formed in the rear surface **142** of the second section **74**, as shown in FIG. **24**. The dowel openings **148** are configured to receive a portion of second alignment dowels **150**, as shown in FIG. **14**. The second alignment dowels **150** are configured to align the second section **74** and the third section **76** during assembly.

[0218] Continuing with FIGS. **14**, **15**, **23** and **24**, a first annular groove **152** is formed in the front surface **140** of the second section **74** such that it surrounds an opening of the horizontal bore **70**, as shown in FIG. **23**. The first groove **152** is positioned between the horizontal bore **70** and the plurality of passages **144** and is configured to receive a first seal **154**, as shown in FIGS. **14** and **15**. Likewise, a second annular groove **156** is formed in the rear surface **142** of the second section **74** and positioned between the horizontal bore **70** and the plurality of passages **144**, as shown in FIG. **24**. The second groove **156** is configured to receive a second seal **158**, as shown in FIGS. **14** and **15**. The seals **154** and **158** shown in FIGS. **14** and **15** are O-rings. The seals **154** and **158** prevent fluid from leaking between the first and second sections **72** and **74** and between the second and third sections **74** and **76** during operation.

[0219] Continuing with FIGS. **14**, **23**, and **24**, a pair of upper and lower suction bores **160** and **162** are formed within the second section **74** and interconnect the intermediate surface **68** and the horizontal bore **70**. The upper and lower suction bores **160** and **162** shown in FIG. **14** are collinear. In alternative embodiments, the bores **160** and **162** may be offset from one another and not collinear.

[0220] Continuing with FIGS. **9** and **14**, the suction bores **160** and **162** are each configured to receive a suction conduit **166**, as shown in FIG. **9**. The suction conduit **166** comprises a first connection member **164** configured to mate with the housing **60**. Each suction bore **160** and **162** opens into a counterbore **168** sized to receive a portion of the first connection member **164**. Internal threads **170** are formed in a portion of the walls of the counterbore **168** for mating with external threads **172**, shown in FIG. **78**, formed on the first connection member **164**.

[0221] Continuing with FIGS. **9** and **14**, a groove **174** is formed in the walls of the counterbore **168** and configured to receive a seal **176**, as shown in FIG. **9**. The seal **176** engages an outer surface of the first connection member **164** to prevent fluid from leaking from the housing **60** during operation. The suction conduits **166** will be described in more detail later herein.

[0222] Continuing with FIGS. **9**, **14**, and **15**, the walls of the horizontal bore **70** within the second section **74** are configured to receive a majority of the fluid routing plug **132**, as shown in FIG. **9**. A small amount of clearance may exist between the walls of the second section **74** and an outer surface of the fluid routing plug **132**.

Third Section of Housing

[0223] Turning to FIGS. **14-16**, **25** and **26**, the third section **76** of the housing **60** is positioned at the rear end of the housing **60** and includes the rear surface **66**. The third section **76** has a generally cylindrical cross-sectional shape. Fluid pressure within the third section **76** varies during operation.

Thus, the third section **76** may be referred to as the dynamic or variable pressure section of the housing **60**.

[0224] Continuing with FIGS. **25** and **26**, the third section **76** comprises a front surface **178** joined to the rear surface **66** of the housing **60** by a portion of the outer intermediate surface **68** and a portion of the horizontal bore **70**. The outer intermediate surface **68** of the third section **76** varies in diameter such that the third section **76** comprises a front portion **180** joined to a rear portion **182**.

[0225] Continuing with FIGS. **15**, **16**, **25**, and **26**, the front portion **180** has a constant outer diameter and has a plurality of passages **184** formed therein. The passages **184** interconnect the front surface **178** and a medial surface **186** of the third section **76**. The passages **184** align with the plurality of passages **144** formed in the second section **74** and the threaded openings **96** formed in the first section **72** of the housing **60**, as shown in FIGS. **14** and **15**. The passages **184** are configured to receive the first fasteners **78** used to secure the sections **72**, **74**, and **76** together.

[0226] Continuing with FIGS. **14** and **25**, a plurality of dowel openings **188** are formed in the front surface **178** of the third section **76**, as shown in FIG. **25**. The dowel openings **188** are configured to receive a portion of second alignment dowels **150**, as shown in FIG. **14**. The second alignment dowels **150** are configured to properly align the third section **76** within the second section **74** during assembly.

[0227] Continuing with FIGS. **25** and **26**, the rear portion **182** of the third section **76** comprises a neck **190** joined to a shoulder **192**. The neck **190** interconnects the front portion **180** and the shoulder **192**. The shoulder **192** includes the rear surface **66** of the housing **60**. The neck **190** has a smaller outer diameter than that of the front portion **180** and the shoulder **192** to provide clearance for the plurality of passages **184** formed in the front portion **180**.

[0228] With reference to FIGS. **25** and **26**, a plurality of first threaded openings **194** are formed in the rear surface **66** of the third section **76**. The first threaded openings **194** are configured to receive a plurality of second fasteners **196**, as shown in FIG. **29**. The second fasteners **196** are configured to secure a stuffing box **198** and a rear retainer **200** to the third section **76** of the housing **60**, as shown in FIG. **29**. The stuffing box **198** and the rear retainer **200** will be described in more detail later herein.

[0229] With reference to FIGS. **26** and **32**, a plurality of second threaded openings **202** are also formed in the rear surface **66** of the third section **76**, as shown in FIG. **26**. The second threaded openings **202** are configured to receive a plurality of third fasteners **204**. The third fasteners **204** are configured to secure a retention plate **206** to the rear surface of the housing **60**, as shown in FIG. **32**. The retention plate **206** will be described in more detail later herein. A plurality of dowel openings **207** are also formed in the rear surface **66** of the third section **76**. The dowel openings **207** are configured to receive third alignment dowels **242**, as shown in FIG. **39**.

[0230] Turning back to FIGS. **9**, **14**, **15**, and **25**, a counterbore **208** is formed in the walls of the horizontal bore **70** within the third section **76** and opens on the front surface **178**. The counterbore **208** is configured to receive a hardened insert **210**, as shown in FIG. **9**. The insert **210** will be described in more detail later herein. The insert **210** engages portions of the fluid routing plug **132** when the fluid routing plug **132** is installed within the housing **60**, as shown in FIG. **9**. The walls of the horizontal bore **70** between the counterbore **208** and the medial surface **186** of the third section **76** are further configured to receive a suction valve guide **212**. A suction valve **214** is also installed within the third section **76** of the housing **60**. The suction valve **214** and suction valve guide **212** will be described in more detail later herein.

[0231] Continuing with FIGS. **9**, **14**, **15**, **25**, and **26**, the walls of the horizontal bore **70** within the neck **190** of the rear portion **182** are sized to receive at least a portion of a reciprocating plunger **216**, as shown in FIG. **9**. The portion of the horizontal bore **70** extending through the neck **190** has a uniform diameter and opens into a first counterbore **218** formed in the shoulder **192**, as shown in FIGS. **14** and **15**. The first counterbore **218** is sized to receive a portion of the stuffing box **198**, as shown in FIG. **9**. The first counterbore **218** opens into a second counterbore **220**, of which opens on

the rear surface **66** of the housing **60**, as shown in FIGS. **14**, **15**, and **26**. The second counterbore **220** is sized to receive a wear ring **222** and a seal **224**, as shown in FIG. **9**. The wear ring **222** and the seal **224** each have an annular shape. When such components are installed within the housing **60**, the wear ring **222** surrounds the seal **224**, and the seal **224** engages an outer surface of the stuffing box **198**.

Assembly of Housing

[0232] Turning to FIGS. **27-29**, the housing **60** is assembled by threading a first end **226** of each of the first fasteners **78** into a corresponding one of the threaded openings **96** formed in the first section **72**. Once installed therein, the first fasteners **78** project from the rear surface **80** of the first section **72**. The second and third sections **74** and **76** may then be slid onto the fasteners **78** projecting from the first section **72** using the corresponding passages **144** and **184**. The first and second alignment dowels **100** and **150** help to further align the sections **72**, **74**, and **76** together during assembly.

[0233] Continuing with FIGS. **27-29**, when the second and third sections **74** and **76** are installed on the fasteners **78**, a second end **228** projects from the medial surface **186** of the third section **76**, as shown in FIG. **29**. A flange nut **230** is installed on the second end **228** and torqued against the medial surface **186**, tightly securing the sections **72**, **74**, and **76** together. When the housing **60** is assembled, a footprint of the rear surface **142** of the second section **74** is entirely within a footprint of the rear surface **80** of the first section **72**, as shown in FIG. **29**.

[0234] Continuing with FIGS. **27-29**, the first fastener **78** shown in the figures is a threaded stud. In alternative embodiments, other types of fasteners known in the art may be used instead of a threaded stud. For example, screws or bolts may be used to secure the sections together. In further alternative embodiments, the nut may comprise the three-piece nut **90**, shown in FIGS. **19-21**.

[0235] Continuing with FIGS. **27-29**, to remove a section **72**, **74**, or **76**, the nut **230** is unthreaded from the second end **228** of each first fastener **78**. The sections **72**, **74**, and **76** may then be pulled apart, as needed. If the first section **72** is being replaced, the first fasteners **78** are also unthreaded from the threaded openings **96**. The components installed within the housing **60** may also be removed, as needed, prior to disassembling the housing **60**.

Components Attached to Rear Surface of Housing

[0236] Turning to FIGS. **29-32**, in addition to the housing **60**, the fluid end section **56** comprises a plurality of components attached to the rear surface **66** of the housing **60**. Such components are configured to receive the plunger **216**. The various components include the retention plate **206**, the stuffing box **198**, and the rear retainer **200**, previously mentioned. The components further comprise a plunger packing **300**, and a packing nut **290**.

Retention Plate

[0237] Continuing with FIGS. **29-32**, the retention plate **206** has a cylindrical cross-sectional shape and is sized to cover the rear surface **66** of the housing **60** and the wear ring **222** and the seal **224**, as shown in FIG. **29**. The retention plate **206** holds the wear ring **222** and the seal **224** within the housing **60** in the event the stuffing box **198** needs to be removed.

[0238] Continuing with FIGS. **29-32**, the retention plate **206** comprises opposed front and rear surface **237** and **238** joined by a central opening **239** formed therein. A plurality of first passages **234** are formed in the retention plate **206** and surround the central opening **239** of the plate **206**. The first passages **234** align with the first threaded openings **194** formed in the rear surface **66** of the housing **60** and are configured to receive the plurality of second fasteners **196**.

[0239] Continuing with FIGS. **30-32**, a plurality of second passages **236** are also formed in the retention plate **206**. The second passages **236** align with the second threaded openings **202** formed in the rear surface **66** of the housing **60** and are configured to receive the third fasteners **204**, as shown in FIG. **32**. A third fastener **204** is threaded into one of the second threaded openings **202** and turned until it sits flush with the rear surface **238** of the retention plate **206**, as shown in FIG. **32**.

[0240] Continuing with FIGS. 30-32, a plurality of dowel openings **240** are formed in the retention plate **206** for receiving third alignment dowels **242**, as shown in FIG. 39. The third alignment dowels **242** assist in properly aligning the retention plate **206** and the stuffing box **198** on the housing **60** during assembly.

[0241] Turning back to FIG. 29, since fluid does not contact the retention plate **206** during operation, the retention plate **206** may be made of a different and less costly material than that of the housing **60** or the stuffing box **198**. For example, the retention plate **206** may be made of alloy steel, while the housing **60** and stuffing box **198** are made of stainless steel.

Stuffing Box

[0242] Turning to FIGS. 29, 33, 34, and 39, the stuffing box **198** comprises opposed front and rear surfaces **244** and **246** joined by an outer intermediate surface **248** and a central passage **250** formed therein. The stuffing box **198** further comprises a front portion **252** joined to a rear portion **254**. The front portion **252** has a smaller outer diameter than the rear portion **254** such that a medial surface **256** is formed between the front and rear surfaces **244** and **246**. The front portion **252** includes the front surface **244** of stuffing box **198**, and the rear portion **254** includes the rear surface **246** of the stuffing box **198**. An internal shoulder **272** is formed within the walls of the central passage **250** within the rear portion **254** of the stuffing box **198**.

[0243] Continuing with FIGS. 33, 34, and 39, a plurality of passages **258** are formed within the rear portion **254** of the stuffing box **198** and interconnect the medial surface **256** and the rear surface **246**. The passages **258** are configured to align with the plurality of first passages **234** formed in the retention plate **206** and the plurality of first threaded openings **194** formed in the rear surface **66** of the housing **60**, as shown in FIG. 29.

[0244] Continuing with FIGS. 33, 34, and 39, a plurality of dowel openings **260** may be formed in the medial surface **256** of the stuffing box **198**. The dowel openings **260** are configured to receive at least a portion of the third alignment dowels **242** to properly align the stuffing box **198** on the retention plate **206** and the housing **60** during assembly, as shown in FIG. 39. Likewise, a plurality of dowel openings **268** may be formed in the rear surface **246** of the stuffing box **198** for receiving fourth alignment dowels **270**, as shown in FIG. 39. The fourth alignment dowels **270** assist in properly aligning the rear retainer **200** on the stuffing box **198** during assembly.

[0245] Continuing with FIG. 39, the stuffing box **198** is installed within the third section **76** of the housing **60** such that the front portion **252** is disposed within the horizontal bore **70** and the medial surface **256** abuts the rear surface **238** of the retention plate **206**. The outer intermediate surface **248** of the front portion **252** of the stuffing box **198** engages the seal **224**. The seal **224** prevents fluid from leaking between the housing **60** and the stuffing box **198**.

[0246] Continuing with FIG. 39, during operation, the seal **224** wears against the outer intermediate surface **248** of the front portion **252**. Should the front portion **252** begin to erode, the stuffing box **198** may be removed and replaced with a new stuffing box **198**. Likewise, the seal **224** wears against the wear ring **222** during operation. The wear ring **222** is preferably made of a harder and more wear resistant material than the housing **60**, such as tungsten carbide. Should the wear ring **222** begin to erode, the wear ring **222** can be removed and replaced with a new wear ring **222**.

Locating the seal **224** between replaceable parts protects the housing **60** over time.

Rear Retainer

[0247] Turning to FIGS. 29, 35, 36, and 39, the rear retainer **200** comprises opposed front and rear surfaces **276** and **278** joined by an outer intermediate surface **280** and a central passage **282** formed therein. A plurality of passages **284** are formed in the rear retainer **200** and surround the central passage **282**. The passages **284** interconnect the front and rear surfaces **276** and **278** of the rear retainer **200** and are configured to align with the passages **258** formed in the rear portion **254** of the stuffing box **198**, as shown in FIG. 29. A plurality of dowel openings **286** are formed in the front surface **276** of the rear retainer **200** for receiving a portion of the fourth alignment dowels **270**, as shown in FIG. 39.

[0248] Continuing with FIGS. 35, 36, 39, and 40, an internal shoulder 279 is formed within the walls of the central passage 282 of the rear retainer 200. Internal threads 288 are formed in the walls of the central passage 282 and positioned between the internal shoulder 279 and the rear surface 278. The internal threads 288 are configured to receive a packing nut 290, as shown in FIG. 39. The walls positioned between the internal shoulder 279 and the front surface 276 are smooth and include one or more lubrication ports 292. The lubrication port 292 interconnects the central passage 282 and the outer intermediate surface 280 of the rear retainer 200, as shown in FIG. 40.

Plunger Packing and Packing Nut

[0249] Continuing with FIG. 39, fluid is prevented from leaking around the plunger 216 during operation by a plunger packing 300. The plunger packing 300 is installed within the stuffing box 198 and comprises a plurality of packing seals 302 sandwiched between first and second metal rings 304 and 306. The first metal ring 304 abuts the internal shoulder 272 formed within the stuffing box 198 and the second metal ring 306 extends into the central passage 282 formed in the rear retainer 200. The second metal ring 306 is known in the art as a “lantern ring”. One or more passages 303, shown in FIG. 29, may be formed in the second metal ring 306 and fluidly connect with the one or more lubrication ports 292 formed in the rear retainer 200. During operation, oil used to lubricate the plunger 216 and plunger packing 300 is supplied through the lubrication port 292 and second metal ring 306.

[0250] With reference to FIGS. 37-39, the plunger packing 300 is retained within the stuffing box 198 and the rear retainer 200 using the packing nut 290. The packing nut 290 comprises opposed front and rear surfaces 308 and 310 joined by an outer intermediate surface 312 and a central passage 314 formed therein. External threads 316 are formed in a portion of the outer intermediate surface 312 for engaging the internal threads 288 formed in the rear retainer 200, as shown in FIG. 39. When the packing nut 290 is installed within the rear retainer 200, the front surface 308 of the packing nut 290 engages and compresses the plunger packing 300, as shown in FIG. 39. When compressed, the packing seals 302 of the plunger packing 300 tightly seal against an outer surface of the plunger 216.

[0251] Continuing with FIG. 39, during operation, the packing nut 290 may be tightened, as needed, to ensure adequate compression of the packing seals 302 against the plunger 216. At least a portion of the packing nut 290 projects from the rear surface 278 of the rear retainer 200 to provide clearance to turn the packing nut 290, as needed. The central passage 314 formed in the packing nut 290 is sized to closely receive the plunger 216. A groove 318 may be formed in the walls of the central passage 314 for receiving a seal 320. The seal 320 shown in FIG. 39 is an O-ring. The seal 320 prevents fluid and lubricant from leaking around the plunger 216 during operation.

Assembly of Components on Rear Surface of Housing

[0252] Turning back to FIG. 29, the front surface 276 of the rear retainer 200 abuts the rear surface 246 of the stuffing box 198 such that the plurality of passages 284 align with the plurality of passages 258 formed in the stuffing box 198. A second fastener 196 is installed within a corresponding one of the aligned first threaded openings 194 and passages 234, 258, and 284. A first end 294 of the second fastener 196 threads into the first threaded opening 194 and a second end 296 projects from the rear surface 278 of the rear retainer 200. A nut 298 is threaded onto the second end 296 and torqued against the rear surface 278, tightly securing the stuffing box 198 and the rear retainer 200 to the third section 76 of the housing 60.

[0253] Continuing with FIG. 29, the nut 298 shown in the figures is a 12-point flange nut. In alternative embodiments, the nut may comprise the three-piece nut 90, shown in FIGS. 19-21. The second fastener 196 shown in the figures is a threaded stud. In alternative embodiments, the second fastener 196 may comprise other fasteners known in the art, such as a bolt or screw.

[0254] Continuing with FIG. 29, the stuffing box 198 and rear retainer 200 are attached to the housing 60 after the retention plate 206 has first been attached to the rear surface 66 of the housing 60. The plunger packing 300 may be installed within stuffing box 198 either before or after the

stuffing box **198** is attached to the housing **60**. After all the components are assembled, the packing nut **290** is threaded into the rear retainer **200** until it engages the plunger packing **300**.

[0255] With reference to FIGS. **29** and **39**, when the retention plate **206**, the stuffing box **198**, and rear retainer **200** are attached to the housing **60**, the central opening **239** of the retention plate **206** and the central passages **250** and **282** of the stuffing box **198** and the rear retainer **200** form an extension of the horizontal bore **70**. Likewise, the interior of the plunger packing **300** and the central passage **314** of the packing nut **290** also form extensions of the horizontal bore **70**. The plunger **216** is installed within the fluid end section **56** through the rear surface **310** of the packing nut **290**. During operation, the plunger **216** reciprocates within the horizontal bore **70**, creating the variance in fluid pressure within the fluid end section **56** during operation.

[0256] With reference to FIGS. **3**, **39**, and **40**, during operation, reciprocal movement of the plunger **216** is driven by a pony rod **322** installed within the power end assembly **54**. A clamp **324** secures the plunger **216** to the pony rod **322** such that the plunger **216** and pony rod **322** move in unison.

Components Installed within the Housing

[0257] Turning to FIGS. **40-79**, the various internal components of the housing **60** will now be described in more detail. Fluid is routed throughout the housing **60** by the fluid routing plug **132**. The timing of movement throughout the fluid routing plug **132** is controlled by the suction valve **214** and the discharge valve **138**. Movement of the valves **214** and **138** is guided by the suction valve guide **212** and the discharge plug **120**.

Fluid Routing Plug

[0258] Turning to FIGS. **40-59**, the fluid routing plug **132** comprises a body **330** having a suction surface **332** and an opposed discharge surface **334** joined by an outer intermediate surface **336**. A central longitudinal axis **338** extends through the body **330** and the suction and discharge surfaces **332** and **334**. When the fluid routing plug **132** is installed within the housing **60**, at least a portion of the discharge surface **334** is positioned within the first section **72** of the housing **60**, and at least a portion of the suction surface **332** is positioned within the third section **76** of the housing **60**, as shown in FIGS. **40** and **41**.

[0259] Continuing with FIGS. **43-59**, the body **330** further comprises a plurality of suction fluid passages **340**. The suction passages **340** interconnect the intermediate surface **336** and the suction surface **332** of the body **330**, as shown in FIG. **48**. The connection is formed within a blind bore **342** formed within the suction surface **332** of the body **330**. The blind bore **342** may be referred to as an axially-blind bore **342** because it is blind along the longitudinal axis **338** of the body **330**. During operation, fluid entering the housing **60** through the suction bores **160** and **162** flows into the suction passages **340** of the fluid routing plug **132** and into the axially-blind bore **342**. From there, fluid flows towards the suction surface **332** of the body **330** and out of the fluid routing plug **132**. Three suction fluid passages **340** are shown in FIGS. **43-59**. In alternative embodiments, more or less than three suction fluid passages **340** may be formed within the body **330**.

[0260] Continuing with FIGS. **49** and **50**, each suction passage **340** has a generally oval or tear drop cross-sectional shape. An opening **344** of each suction passage **340** on the intermediate surface **336** comprises a first side wall **346** joined to a second side wall **348** by first and second ends **350** and **352**. The first and second side walls **346** and **348** are straight lines of equal length S , and the first and second ends **350** and **352** are circular arcs, as shown in FIG. **50**.

[0261] Continuing with FIG. **50**, the first end **350** of the opening **344** has a radius of $R1$ with a center at $C1$, and the second end **352** has a radius of $R2$ with a center at $C2$. The first end **350** is larger than the second end **352** such that $R1 > R2$. The first and second side walls **346** and **348** are tangent to the first and second ends **350** and **352** and have an included angle, σ .

[0262] Continuing with FIG. **50**, the opening **344** has a centerline **354** that connects the centers $C1$ and $C2$ of the first and second ends **350** and **352**. The centerline **354** has a length E and is parallel with the central longitudinal axis **338**. A cross-sectional shape of each suction passage **340** throughout the length of the body **330** corresponds with the shape of each opening **344**, as shown in

FIG. 55. Each suction passage 340 is sized and shaped to maximize fluid flow through the passage 340 and minimize fluid turbulence and stress to the body 330 of the fluid routing plug 132.

[0263] With reference to FIGS. 55 and 56, each suction fluid passage 340 extends between the axially-blind bore 342 and the suction surface 332 such that each suction passage 340 comprises a longitudinal axis 356. The longitudinal axis 356 extends through the center C1 of the first end 350 of the opening 344 and intersects the central longitudinal axis 338, as shown in FIG. 56.

[0264] Continuing with FIGS. 43-59, the body 330 further comprises a plurality of discharge fluid passages 360. The discharge passages 360 interconnect the suction surface 332 and the discharge surface 334 of the body 330 and do not intersect any of the suction passages 340. Rather, the discharge and suction passages 360 and 340 are in a spaced-relationship. In operation, fluid exiting the body 330 at the suction surface 332 is subsequently forced into the discharge passages 360, towards the discharge surface 334 of the body 330, and out of the fluid routing plug 132. Three discharge fluid passages 360 are shown in FIGS. 43-59. In alternative embodiments, more or less than three discharge fluid passages 360 may be formed within the body 330.

[0265] Continuing with FIGS. 43, 46, and 48, the suction surface 332 of the body 330 comprises an outer rim 362 joined to the axially-blind bore 342 by a tapered seating surface 366, as shown in FIGS. 46 and 48. Likewise, the discharge surface 334 comprises an outer rim 368 joined to a central base 370 by a tapered seating surface 372, as shown in FIGS. 43 and 48.

[0266] Continuing with FIGS. 43, 46, and 48, each discharge passage 360 opens at a first opening 374 on the outer rim 362 of the suction surface 332 and opens at a second opening 376 on the central base 370 of the discharge surface 334. The second openings 376 surround a blind bore 378 formed in the central base 370 of the discharge surface 334. The blind bore 378 is configured to engage a tool used to grip the fluid routing plug 132, as needed. For example, the walls of the blind bore 378 may be threaded. The central base 370 may also be slightly recessed from the tapered seating surface 372 such that a small counterbore 380 is created. The counterbore 380 helps further reduce any turbulence of fluid exiting the second openings 376.

[0267] Continuing with FIG. 54, a position of the first and second openings 374 and 376 of each discharge passage 360 may be determined relative to a plane containing a line 382 that is perpendicular to the central longitudinal axis 338. The first opening 374, when projected onto the plane, is positioned at a first distance F1 from the central longitudinal axis 338 and at a first angle $\phi 1$ relative to the line 382. The second opening 376, when projected onto the plane, is positioned at a second distance F2 from the central longitudinal axis 338 and at a second angle $\phi 2$ relative to the line 382.

[0268] The first and second distances F1 and F2 shown in FIG. 54 are different. Likewise, the first and second angles $\phi 1$ and $\phi 2$ shown in FIG. 54 are different. In alternative embodiments, the first and second angles $\phi 1$ and $\phi 2$ may be different, but the first and second distances F1 and F2 may be the same. In further alternative embodiments, the first and second angles $\phi 1$ and $\phi 2$ may be the same, but the first and second distances F1 and F2 may be different. In even further alternative embodiments, the first and second distances F1 and F2 may be the same, and the first and second angles $\phi 1$ and $\phi 2$ may be the same.

[0269] With reference to FIGS. 51-53 and 56-59, each discharge passage 360 has an arced cross-sectional shape. The length of the arc may gradually increase between the suction and discharge surfaces 332 and 334, as shown in FIGS. 51-53. In alternative embodiments, the discharge passages 360 may have different shapes and sizes.

[0270] Turning back to FIGS. 44 and 48, a first annular groove 384 is formed in the outer intermediate surface 336 of the body 330 for housing the first seal 386. The first groove 384 is positioned adjacent the discharge surface 334 and is characterized by two side walls 388 joined by a base 390. When the fluid routing plug 132 is installed within the housing 60, the first seal 386 engages an inner surface of the wear ring 136 installed within the first section 72 of the housing 60, as shown in FIGS. 9, 40, and 41. During operation, the first seal 386 wears against the wear ring

136. If the wear ring **136** begins to erode, the wear ring **136** may be removed and replaced with a new wear ring **136**. The wear ring **136** has an annular shape and may be made of a harder and more wear resistant material than the housing **60**. For example, the housing **60** may be made of stainless steel and the wear ring **136** made of tungsten carbide.

[0271] With reference to FIGS. **42**, **44**, and **48**, a second annular groove **392** is formed in the outer intermediate surface **336** of the body **330** for housing a second seal **394**. The second groove **392** is positioned adjacent the suction surface **332** and is characterized by a plurality of side walls **396** joined by a base **398**, as shown in FIG. **42**. Four side walls **396** are shown in FIG. **42** such that the groove **392** has a rounded shape. When the fluid routing plug **132** is installed within the housing **60**, the second seal **394** engages an inner intermediate surface **420** of the hardened insert **210**, as shown in FIG. **42**. During operation, the second seal **394** wears against the insert **210**. If the insert **210** begins to erode, the insert **210** may be removed and replaced with a new insert **210**.

[0272] Continuing with FIGS. **42**, **44**, and **48**, the outer intermediate surface **336** of the body **330** further comprises an annular shoulder **400** formed in the body **330**. The shoulder **400** is positioned between the opening **344** of the suction passages **340** and the second groove **392**. When the fluid routing plug **132** is installed within the housing **60**, the shoulder **400** abuts a front surface **416** of the insert **210**, as shown in FIG. **42**. Axial movement of the fluid routing plug **132** towards the rear surface **66** of the housing **60** is prevented by the engagement between the shoulder **400** and the insert **210**. During operation, the shoulder **400** may wear against the insert **210**. If either feature begins to wear, the fluid routing plug **132** and/or the insert **210** may be removed and replaced with a new fluid routing plug **132** and/or insert **210**.

[0273] Continuing with FIGS. **40**, **41**, **44**, and **48**, the outer intermediate surface **336** of the body **330** adjacent the first groove **384** is characterized as a first cylindrical surface **404**. Likewise, the outer intermediate surface **336** adjacent the annular shoulder **400** is characterized as a second cylindrical surface **406**. The first cylindrical surface **404** has a maximum outer diameter that is equal or almost equal to a maximum outer diameter of the second cylindrical surface **406**. The surfaces **404** and **406** are configured to closely face the walls of the horizontal bore **70** within the second section **74** of the housing **60**, as shown in FIGS. **40** and **41**.

[0274] Continuing with FIGS. **40**, **41**, **44**, **48**, and **49**, the outer intermediate surface **336** of the body **330** further comprises a first bevel **408** joined to a transition surface **410** formed in the body **330**. The first bevel **408** and the transition surface **410** are positioned between the first cylindrical surface **404** and the openings **344** of the suction passages **340**. The outer intermediate surface **336** of the body **330** slowly tapers outward from the transition surface **410** to the second cylindrical surface **406**.

[0275] Continuing with FIGS. **40** and **41**, when the fluid routing plug **132** is installed within the housing **60**, the first bevel **408** provides clearance between the outer intermediate surface **336** of the fluid routing plug **132** and an opening of the suction bores **160** and **162**. Such clearance gives way to an annular fluid channel **412** formed between the housing **60** and the fluid routing plug **132**. The shape of the outer intermediate surface **336** of the fluid routing plug **132** between the first and second cylindrical surfaces **404** and **406** helps direct fluid flowing from the suction bores **160** and **162** into the openings **344** of the suction passages **340** while minimizing fluid turbulence.

[0276] Turning back to FIG. **42**, the outer intermediate surface **336** of the body **330** further comprises a second bevel **414** formed in the body **330**. The second bevel **414** is positioned between the suction surface **332** and the second groove **392**. The second bevel **414** provides clearance to help install the fluid routing plug **132** within the housing **60** and the insert **210**.

Hardened Insert

[0277] With reference to FIGS. **42** and **60-63**, the insert **210** has an annular shape and comprises opposed front and rear surfaces **416** and **418** joined by inner and outer intermediate surfaces **420** and **422**. The insert **210** further comprises a first bevel **426** formed in the inner intermediate surface **420** adjacent the front surface **416**, as shown in FIG. **63**. The first bevel **426** provides clearance to

assist in installing the fluid routing plug **132** within the insert **210** within the housing **60**, as shown in FIG. **42**. The insert **210** also comprises a second bevel **424** formed in the outer intermediate surface **422** adjacent the rear surface **418**. The second bevel **424** provides clearance to assist in installing the insert **210** within the counterbore **208** formed in the third section **76** of the housing **60**, as shown in FIG. **42**. The insert **210** is made of a harder and more wear resistant material than the housing **60**. For example, if the housing **60** is made of stainless steel, the insert **210** may be made of tungsten carbide.

Suction and Discharge Valves

[0278] With reference to FIGS. **40**, **41**, and **64-67**, the flow of fluid throughout the housing **60** and the fluid routing plug **132** is regulated by the suction and discharge valves **214** and **138**. The suction valve **214** is configured to engage the suction surface **332**, and the discharge valve **138** is configured to engage the discharge surface **334** of the fluid routing plug **132** such that the surfaces **332** and **334** function as valve seats. The valves **214** and **138** are similar in shape but may vary in size. As shown in FIGS. **40** and **41**, the discharge valve **138** is slightly larger than suction valve **214**.

[0279] Continuing with FIGS. **64-67**, the discharge valve **138** is shown in more detail. The suction valve **214** has the same features as the discharge valve **138** so only the discharge valve **138** is shown in more detail in the figures. The discharge valve **138** comprises a stem **402** joined to a body **428**. The body **428** comprises an outer rim **430** joined to a valve insert **432** by a tapered seating surface **434**. An annular cutout **436** formed within the seating surface **434** is configured to house a seal **438**, as shown in FIG. **67**.

[0280] Continuing with FIGS. **40** and **41**, during operation, the seating surface **434** and the seal **438** engage the seating surface **372** of the discharge surface **334** and block fluid from entering or exiting the discharge passages **360**, as shown in FIG. **40**. Likewise, the seating surface **434** and the seal **438** on the suction valve **214** engage the seating surface **366** of the suction surface **332** and block fluid from entering or exiting the suction passages **340**, as shown in FIG. **41**.

[0281] Continuing with FIGS. **40** and **41**, when the seating surfaces **434** and **372** are engaged, the valve insert **432** extends partially into the counterbore **380** formed in the discharge surface **334**. Fluid exiting the second openings **376** of the discharge passages **360** contacts the insert **432**, pushing the discharge valve **138** away from the discharge surface **334** before flowing around the seating surface **434** of the discharge valve **138**. Such motion enlarges the area for fluid to flow between the seating surfaces **372** and **434** before fluid reaches the surfaces **372** and **434**, thereby reducing the velocity of fluid flow within such area. The lowered fluid velocity between the surfaces **372** and **434** causes any wear to the valve **138** or **214** to be concentrated at the insert **432** instead of the crucial sealing elements, thereby extending the life of the valve **138** or **214**.

[0282] Likewise, the insert **432** on the suction valve **214** extends partially into the opening of the axially-blind bore **342**. Fluid within the axially-blind bore **342** contacts the insert **432** before flowing around the seating surface **434** and seal **438** of the suction valve **214**. Such motion enlarges the area for fluid to flow between the seating surfaces **366** and **434** before fluid reaches the surfaces **366** and **434**, thereby reducing the velocity of fluid flow within such area.

[0283] Continuing with FIGS. **64-67**, the stem **402** projects from a top surface **440** of the body **428** of the valve **138** or **214**. The outer rim **430** surrounds the stem **402** and is spaced from the stem **402** by an annular void **442**. A groove **444** is formed in the outer rim **430** for receiving a portion of a spring **446**, as shown in FIGS. **40** and **41**.

[0284] Continuing with FIGS. **40** and **41**, during operation, the valves **138** and **214** move axially along the longitudinal axis **62** of the housing **60** between open and closed positions. In the closed position, the seating surface **434** and the seal **438** of each of the valves **138** and **214** tightly engage the corresponding seating surface **372** or **366** of the fluid routing plug **132** and the valve insert **432** is disposed within the corresponding bore **380** or **342**. In the open position, the seating surface **434** and the seal **438** are spaced from the corresponding seating surface **372** or **366** of the fluid routing

plug **132** and the valve insert **432** is spaced from the corresponding counterbore **380** or bore **342**.
Suction Valve Guide

[0285] With reference to FIGS. **40**, **41**, and **68-71**, axial movement of the suction valve **214** is guided by the suction valve guide **212**. The suction valve guide **212** comprises a thin-walled skirt **448** joined to a body **450** by a plurality of support arms **452**. The skirt **448** comprises a tapered upper section **454** joined to a cylindrical lower section **456**. The plurality of arms **452** join the tapered upper section **454** to the body **450**. A plurality of flow ports **458** are formed between adjacent arms **452** such that fluid may pass through the suction valve guide **212** during operation.

[0286] Continuing with FIGS. **40** and **41**, the suction valve guide **212** is installed within the housing **60** such that the tapered upper section **454** engages a tapered surface **455** of the walls of the horizontal bore **70**. Such engagement prevents further axial movement of the suction valve guide **212** within the housing **60**. When the suction valve guide **212** is installed within the housing **60**, the skirt **448** covers the walls of the housing **60** positioned between the flow ports **458** and the fluid routing plug **132**. During operation, fluid wears against the skirt **448**, thereby protecting the housing **60** from wear and erosion. If the skirt **448** begins to erode, the suction valve guide **212** can be removed and replaced with a new guide **212**.

[0287] Continuing with FIGS. **40**, **41**, and **68-71**, the body **450** of the suction valve guide **212** is tubular and is centered within the skirt **448**. A tubular insert **460** is installed within the body **450**, as shown in FIG. **71**. The insert **460** is configured to receive the stem **402** of the suction valve **214**, as shown in FIGS. **40** and **41**. During operation, the stem **402** moves axially within the insert **460** and wears against the insert **460**. An annular cutout **462** formed in the stem **402**, shown in FIGS. **66** and **67**, provides space for any fluid or other material trapped between the stem **402** and the insert **460**. The insert **460** is made of a harder and more wear resistant material than the body **450** thereby extending the life of the suction valve guide **212**. For example, if the body **450** is made of stainless steel, the insert **460** may be made of tungsten carbide.

[0288] Continuing with FIGS. **40** and **41**, a spring **446** is positioned between the outer rim **430** of the suction valve **214** and the plurality of arms **452** such that the spring **446** surrounds at least a portion of the body **450** of the suction valve guide **212**. During operation, the spring **446** biases the suction valve **212** in a closed position, as shown in FIG. **41**. Fluid pushing against the valve insert **432** moves the suction valve **214** axially to compress the spring **446** and move the suction valve **214** to an open position, as shown in FIG. **40**.

Discharge Plug

[0289] With reference to FIGS. **40**, **41**, and **72-75**, axial movement of the discharge valve **138** is guided by the discharge plug **120**. The discharge plug **120** comprises a pair of legs **464** joined to a body **466**. The body **466** comprises a front portion **468** joined to a rear portion **470** by a medial portion **472**. The medial portion **472** has a larger outer diameter than both the front and rear portions **468** and **470**. An outer surface of the medial portion **472** engages the seal **126** installed within the first section **72** of the housing **60**, as shown in FIGS. **40** and **41**. The pair of legs **464** are joined to the medial portion **472** and extend between the medial portion **472** and the discharge surface **334** of the fluid routing plug **132**.

[0290] Continuing with FIGS. **40**, **41**, and **72-75**, a dowel opening **474** is formed in the outer surface of the medial portion **472** for receiving the locating dowel pin **130**. The discharge plug **120** is installed within the first section **72** of the housing **60** such that the locating dowel pin **130** is installed within the dowel opening **474** formed in the medial portion **472** and the locating cutout **128** formed in the first section **72** of the housing **60**. Such installation aligns the discharge plug **120** within the housing **60** so that the pair of legs **464** do not block the openings of the upper and lower discharge bores **102** and **104**.

[0291] Continuing with FIGS. **40** and **41**, the locating cutout **128** may be large enough to provide sufficient clearance for installation of the locating dowel pin **130** within the locating cutout **128**. The locating cutout **128** is sized to allow maximum clearance for assembly, but still maintain an

acceptable rotational position of the discharge plug **120**. For example, the cutout **128** may be a maximum of 15 degrees wide along the circumference of the horizontal bore **70**.

[0292] Continuing with FIG. **75**, an axially-blind bore **476** extends within the body **466** and opens on the rear portion **470** of the body **466**. The bore **476** is sized to receive a tubular insert **478**. The tubular insert **478** is similar to the tubular insert **460** installed within the suction valve guide **212**. The tubular insert **478** is configured to receive the stem **402** of the discharge valve **138**, as shown in FIGS. **40** and **41**.

[0293] Continuing with FIGS. **40**, **41**, and **75**, during operation, the stem **402** moves axially within the tubular insert **478**. A plurality of passages **480** are formed in the body **466** and interconnect the bore **476** and an outer surface of the medial portion **472**. During operation, any fluid or other material trapped within the bore **476** exits the discharge plug **120** through the passages **480**. A spring **446** is positioned between the medial portion **472** of the plug **120** and the outer rim **430** of the discharge valve **138**, as shown in FIGS. **40** and **41**. The spring **446** biases the discharge valve **138** in the closed position, as shown in FIG. **40**. Fluid pushing against the valve insert **432** moves the discharge valve **138** axially to compress the spring **446** and move the discharge valve **138** to an open position, as shown in FIG. **41**.

[0294] Continuing with FIGS. **40**, **41**, **72**, and **75**, the front portion **468** of the body **466** is sized to be disposed within a counterbore **482** formed within the front retainer **118**. When disposed therein, a rear surface **484** of the front retainer **118** abuts an outer surface of the medial portion **472** of the discharge plug **120**, as shown in FIGS. **40** and **42**. Such engagement holds the discharge plug **120** in place between the front retainer **118** and the fluid routing plug **132**. A blind bore **486** is formed in an outer surface of the front portion **468** of the plug **120**. The blind bore **486** is configured to engage a tool used to help install or remove the plug **120** from the housing **60**. For example, the bore **486** may have threaded walls.

Front Retainer

[0295] With reference to FIGS. **40**, **41**, **76**, and **77**, the front retainer **118** comprises opposed front and rear surfaces **488** and **484** joined by an outer surface having external threads **124** and a horizontal bore **490** formed therein. The horizontal bore **490** comprises a hex portion **492** that opens in the counterbore **482**, as shown in FIGS. **40** and **41**. The hex portion **492** is configured to mate with a tool used to thread the front retainer **118** into the housing **60** until it abuts the discharge plug **120**, as shown in FIGS. **40** and **41**. An annular void **494** is formed within the front surface **488** of the front retainer **118**. The annular void **494** decreases the weight of the front retainer **118**, making it easier to thread into the housing **60**.

Discharge Conduits and Manifold

[0296] With reference to FIG. **41**, each discharge fitting **108** comprises a support base **502** and a connection end **512**. A discharge fitting adapter **504** is installed within the counterbore **106** formed in the upper and lower discharge bores **102** and **104**. When installed, the seal **112** engages an outer surface of the fitting adapter **504**. A groove **505** is formed within the discharge fitting **108** for receiving a second seal **507**. The second seal **507** likewise engages an outer surface of the fitting adapter **504**.

[0297] Continuing with FIG. **41**, the support base **502** is sized to abut the outer intermediate surface **68** of the first section **72** of the housing **60**. The support base **502** comprises a plurality of passages **506**, shown in FIG. **29**, configured to align with the threaded openings **114** formed in the intermediate surface **68** and surrounding the discharge bores **102** and **104**. The threaded fasteners **116** are installed within the aligned passages **506** and openings **114** and tightened to secure the discharge fitting **108** to the first section **72**.

[0298] With reference to FIGS. **3** and **41**, the connection end **512** of the discharge fitting **108** is configured to mate within one or more discharge conduits **500** included in an upper or lower discharge manifold **514** or **516**, as shown in FIG. **3**. The upper and lower discharge manifolds **514** and **516** are supported on rack **518**, as shown in FIG. **3**. The fluid end assembly **52** is disposed

within the interior open area of the rack **518**. The rack **518** supports the upper and lower discharge manifolds **514** and **516** in a spaced position from the discharge bores **102** and **104**. As a result, each discharge conduit **500** has an angled or bent shape. In operation, fluid discharges from the housing **60** through upper and lower discharge bores **102** and **104** and is carried to the corresponding upper or lower discharge manifolds **514** or **516** by the discharge fittings and conduits **108** and **500**.

Suction Conduits and Manifold

[0299] With reference to FIGS. **41** and **78**, each suction conduit **166** comprises the first connection member **164** joined to a second connection member **520** by threads, as shown in FIG. **78**. The first and second connection members **164** and **520** may be made of a metal or hardened material.

[0300] Continuing with FIG. **78**, the first connection member **164** comprises an upper portion **524** joined to a lower portion **526**. External threads **172** are formed on a portion of the lower portion **526** for mating with the internal threads **170** formed in the suction bores **160** or **162**. The seal **176** installed within the housing **60** engages a cylindrical outer surface of the lower portion **526** below the external threads **172**. The upper portion **524** has a larger outer diameter than the lower portion **526** and is positioned outside of the housing **60**. The lower portion **526** may abut the counterbore **168** of the suction bores **160** and **162** of the second section **74** of the housing **60**.

[0301] With reference to FIGS. **3** and **78**, the second connection member **520** is configured to mate with one or more connection members or hoses **528** formed on an upper or lower suction manifold **530** or **532**. The upper and lower suction manifolds **530** and **532** are supported on the rack **518** adjacent the discharge manifolds **514** and **516**. The connection members or hoses **528** may be flexible so that they may bend, as needed, to properly interconnect the suction conduits **166** and the suction manifolds **530** and **532**. In operation, fluid is drawn into the housing **60** from the suction manifolds **530** and **532** via the connection members **528**, the suction conduits **166**, and the upper and lower suction bores **160** and **162**.

Assembly of Fluid End Section and Assembly

[0302] Turning to FIGS. **9**, **29**, **79**, and **80**, prior to assembling the housing **60**, the wear ring **136** is preferably first pressed into the counterbore **134** formed in the first section **72** of the housing **60**. Likewise, the hardened insert **210** is pressed into the counterbore **208** formed in the third section **76** of the housing **60**. The seals **126**, **112**, and **176** may also be installed within the first and second sections **72** and **74** of the housing **60**. The wear ring **222** and seal **224** may also be installed within the third section **76** of the housing **60** prior to assembling the housing **60**.

[0303] Following installation of the above described components, the housing **60** may be assembled as described above. Thereafter, the retention plates **206**, stuffing box **198**, rear retainer **200**, plunger packing **300**, and packing nut **290** may be attached to the rear surface **66** of the housing **60**. The inner components of the housing **60** are inserted within the housing **60** through the front surface **64** of the first section **72**. The inner components may be installed prior to attaching the components to the rear surface **66** of the housing **60**, if desired. Following assembly of each fluid end section **56**, each section **56** is attached to the power end assembly **54** using the stay rods **58**.

[0304] Each fluid end section **56** and its various components are heavy and cumbersome. Various tools or lifting mechanisms may be used to assemble the fluid end assembly **52** and attach it to the power end assembly **54**, creating the high pressure pump **50**.

Operation of Fluid End Assembly

[0305] Turning back to FIGS. **40** and **41**, in operation, retraction of the plunger **216** out of the housing **60** pulls fluid from the upper and lower suction bores **160** and **162** into the suction passages **340** within the fluid routing plug **132**. Fluid flowing through the suction passages **340** and into the axially-blind bore **342** pushes on the valve insert **432** of the suction valve **214**, causing the valve **214** to compress the spring **446** and move to an open position, as shown in FIG. **40**. When in the open position, fluid flows around the suction valve **214** and the suction valve guide **212** and into the open horizontal bore **70** within the third section **76** of the housing **60**.

[0306] Continuing with FIG. **41**, extension of the plunger **216** further into the housing **60** pushes

against fluid within the open horizontal bore **70** and forces the fluid towards the suction surface **332** of the fluid routing plug **132**. Such motion also causes the suction valve **214** to move to a closed position, sealing the opening of the axially-blind bore **342**. Because the bore **342** is sealed, fluid is forced into the discharge passages **360**.

[0307] Fluid flowing through the discharge passages **360** contacts the valve insert **432** on the discharge valve **138**, causing the discharge valve **138** to compress the spring **446** and move into an open position, as shown in FIG. **41**. When in the open position, fluid flows around the discharge valve **138** and into the upper and lower discharge bores **102** and **104**. Because fluid exiting the discharge passages **360** has been compressed by extension of the plunger **216** into the housing **60**, such fluid has a higher fluid pressure than that entering the housing **60** through the suction bores **160** and **162**.

[0308] During operation, the plunger **216** continually reciprocates within the housing **60**, pressuring all fluid drawn into the housing **60** through the suction bores **160** and **162**. Pressurized fluid exiting the housing **60** through the upper and lower discharge bores **102** and **104** is delivered to the upper and lower discharge manifolds **514** and **516** in communication with each of the fluid end sections **56**. Pressurized fluid within the discharge manifolds **514** and **516** is eventually delivered to the wellhead **18**, as shown in FIG. **2**.

ALTERNATIVE EMBODIMENTS

[0309] Turning to FIGS. **81-124**, alternative embodiments of fluid end sections that may be used to assemble a fluid end assembly, like the fluid end assembly **52** shown in FIG. **7**, are shown. For ease of reference, those components that are the same or nearly the same as components making up the fluid end section **56** shown in FIG. **9** will be given the same reference numbers.

Fluid End Section **600**

[0310] Turning to FIGS. **81-88**, another embodiment of a fluid end section **600** is shown. The fluid end section **600** is generally identical to the fluid end section **56**, with a few exceptions. The fluid end section **600** comprises a housing **602** having opposed front and rear surfaces **604** and **606** joined by an outer intermediate surface **608** and a horizontal bore **610** formed therein, as shown in FIG. **86**. The housing **602** is identical to the housing **60**, shown in FIGS. **10** and **11**, but comprises another embodiment of a second section **612** of the housing **602**.

[0311] Continuing with FIGS. **81-88**, the second section **612** of the housing **602** comprises opposed front and rear surfaces **614** and **616** joined by a portion of the outer intermediate surface **608** and a portion of the horizontal bore **610**, as shown in FIG. **86**. The second section **612** has the shape of a rectangular prism and is the same height and width as the rear surface **80** of the first section **72**.

[0312] Continuing with FIGS. **81-85**, a plurality of first passages **618** are formed in the second section **612**. Each passage **618** interconnects the front and rear surfaces **614** and **616** and is configured to align with the plurality of passages **86** formed in the first section **72** of the housing **602**, as shown in FIG. **85**. Each passage **618** is configured to each receive a corresponding one of the stay rods **58**. The stay rods **58** pass through the aligned passages **86** and **618**, such that both the first and second sections **72** and **612** of the housing **602** are attached to the power end assembly **54** using the stay rods **58**.

[0313] Continuing with FIGS. **83-85**, in contrast to the fluid end section **56**, each sleeve **94** surrounding a corresponding stay rod **58** abuts the second section **612** of the housing **602**, instead of the first section **72**. Because the sleeves **94** abut the second section **612**, the passages **86** formed in the first section **72** do not include the counterbore **87** for receiving the dowel sleeve **93**, as shown in FIG. **22**. Instead, each passage **618** includes a counterbore **620** that opens on the rear surface **616** of the second section **612** for receiving a corresponding one of the dowel sleeves **93**, as shown in FIG. **85**.

[0314] Continuing with FIG. **86**, a plurality of second passages **622** are also formed in the second section **612**. The second passages **622** surround the horizontal bore **610** and are configured to align with the threaded openings **96** formed in the first section **72** and the first passages **184** formed in

the third section **76**. The first fasteners **78** are installed within the aligned threaded openings **96** and passages **622** and **184** to secure the first, second, and third sections **72**, **612**, and **76** of the housing **602** together.

[0315] Continuing with FIG. **88**, a plurality of dowel openings **624** may be formed in the rear surface **616** of the second section **612** for receiving the second alignment dowels **150**. The first alignment dowels **100** may not be used between the first and second sections **72** and **612** since the stay rods **58** extend through both sections **72** and **612**. However, the sections **72** and **612** may be configured to receive the first alignment dowels **100**, if desired.

[0316] Continuing with FIG. **88**, the walls of the horizontal bore **610** within the second section **612** of the housing **602** are configured to receive another embodiment of a fluid routing plug **630**. The fluid routing plug **630** is generally identical to the fluid routing plug **132**, shown in FIGS. **43** and **44**, but the fluid routing plug **630** comprises another embodiment of an outer intermediate surface **632**.

[0317] With reference to FIGS. **88-94**, a first groove **634** configured to receive the first seal **386** is formed in the intermediate surface **632** of the fluid routing plug **630** adjacent a discharge surface **636**. The first seal **386** engages the wear ring **136** installed within the first section **72**. Immediately adjacent the first groove **634**, the intermediate surface **632** further comprises a cylindrical section **638** joined to landing bevel **640** and a transition section **641**, as shown in FIG. **94**. The cylindrical section **638** and the landing bevel **640** are configured to engage a landing bevel counterbore **642** formed in the second section **612** of the housing **602**, as shown in FIG. **93**.

[0318] Continuing with FIG. **93**, the landing bevel **640** is configured so as to only contact the landing bevel counterbore **642** at a single point or at a very small engagement length. The cross-sectional profile of the landing bevel **640** is thus characterized as a splined curve. In operation, the varying fluid pressures in and around the fluid routing plug **630** cause the plug **630** to stretch or deform as the plunger **216** reciprocates. The landing bevel **640** and landing bevel counterbore **642** are sized so that the surfaces more fully engage as the fluid routing plug **630** stretches or deforms during operation. Such engagement supports the housing **602** and fluid routing plug **630** when in their highest state of loading, thereby reducing stress and increasing the life span of such components.

[0319] Continuing with FIGS. **88-92**, in contrast to the fluid routing plug **132**, the fluid routing plug **630** does not include a second groove **392** for receiving a second seal or an annular shoulder **400**. Instead, the intermediate surface **632** adjacent a suction surface **644** comprises a sealing surface **646**. The sealing surface **646** is configured to engage a second seal **648** installed within the counterbore **208** formed in the third section **76**, as shown in FIG. **88**.

[0320] Continuing with FIG. **88**, in contrast to the fluid end section **56**, the insert **210** is not installed within the counterbore **208**. Rather, the second seal **648** and a second wear ring **650** are installed within the counterbore **208**. Since the fluid routing plug **630** does not engage the insert **210**, engagement of the landing bevel **640** with the landing bevel counterbore **642** prevents further axial movement of the fluid routing plug **630** within the housing **602** during installation. The fluid routing plug **630** is described in more detail in U.S. Pat. No. 11,686,296, authored by Son et al., and issued on Jun. 7, 2023, the entire contents of which are incorporated herein by reference (hereinafter “the '736 application”).

Fluid End Section **700**

[0321] Turning to FIGS. **95-102**, another embodiment of a fluid end section **700** is shown. The fluid end section **700** is similar to the fluid end section **600**, with a few exceptions. The fluid end section **700** comprises a housing **702** having opposed front and rear surfaces **704** and **706** joined by an outer intermediate surface **708** and a horizontal bore **710** formed therein, as shown in FIG. **102**. In contrast to the housing **602**, the housing **702** only comprises two sections—an integral first and second section **712** joined to the third section **76**.

[0322] Continuing with FIGS. **95-102**, the integral first and second section **712** is generally

identical to the first section **72** and the second section **612** of the housing **602**, but the sections are one-piece. The stay rods **58** extend through the entire integral section **712**, as shown in FIG. **101**. The integral section **712** has the shape of a rectangular prism and comprises opposed front and rear surfaces **714** and **716**. A plurality of passages **718** are formed in the integral section **712** for receiving the stay rods **58**. Each passage **718** interconnects the front and rear surfaces **714** and **716** of the integral section **712** and includes a counterbore **720** that opens on the rear surface **716**. A dowel sleeve **93** is installed within each counterbore **720**. The first end **84** of each stay rod **58** projects from the front surface **714** of the integral section **712**. While not shown, the front surface **714** of the integral section **712** may include the notches **82** for receiving the first end **84** of each of the stay rods **58**.

[0323] Continuing with FIGS. **98** and **101**, the rear surface **716** of the integral section **712** comprise a plurality of threaded openings **722** and dowel pin openings **726**. The threaded openings **722** align with the passages **184** formed in the third section **76**, as shown in FIG. **101**. The aligned openings **722** and passages **184** are configured to receive a plurality of first fasteners **724** configured to secure the sections **712** and **76** together. The first fasteners **724** are identical to the first fasteners **78**, but the first fasteners **724** may be shorter in length. The plurality of dowel openings **726** may be formed in the rear surface **716** of the integral section **712** for receiving the second alignment dowels **150**.

[0324] Continuing with FIG. **102**, the components installed within the housing **702** are identical to those described with reference to the fluid end section **56**, with the exception that another embodiment of a fluid routing plug **730** is shown installed therein. The fluid routing plug **730** is generally identical to the fluid routing plug **132**, but the fluid routing plug **730** comprises another embodiment of an outer intermediate surface **732**.

[0325] Continuing with FIGS. **103-108**, the outer intermediate surface **732** comprises a first sealing surface **734** positioned adjacent a discharge surface **736** and a second sealing surface **738** positioned adjacent a suction surface **740**. The first sealing surface **734** engages a first seal **742** installed within a first groove **744** formed in the integral section **712**. In alternative embodiments, the first seal **742** may be surrounded by a wear ring also installed within the first groove **744**. The second sealing surface **738** engages the second seal **648** installed within the counterbore **208** formed in the third section **76**.

[0326] Continuing with FIGS. **103-108**, the outer intermediate surface **732** further comprises a first bevel **746** positioned between the first sealing surface **734** and suction passages **748**, and a second bevel **750** positioned between the suction passages **748** and the second sealing surface **738**. The first bevel **746** engages a beveled surface **752** formed in the walls of the integral section **712**, as shown in FIG. **103**. The second bevel **750** faces a second beveled surface **754** formed in the walls of the integral section **712**, as shown in FIG. **103**. During operation, the second bevel **750** may engage the second beveled surface **754** as the fluid routing plug **730** stretches or deforms. The fluid routing plug **730** is described in more detail in the '736 application.

[0327] Turning back to FIGS. **95** and **96**, the fluid end section **700** further comprises alternative embodiments of a front and rear retainer **760** and **762**. The front retainer **760** is identical to the front retainer **118**, but the front retainer **760** does not include the annular void **494**, shown in FIG. **76**. In alternative embodiments, the front retainer **760** may include an annular void. The rear retainer **762** is identical to the rear retainer **200**, shown in FIG. **36**, but a plurality of passages **764** formed in the rear retainer **762** include a counterbore **766**, as shown in FIG. **96**. The first end of each second fastener **196** and the corresponding nut **298** installed thereon is positioned within a corresponding one of the counterbores **766**.

Fluid End Section **800**

[0328] Turning to FIGS. **109-116**, another embodiment of a fluid end section **800** is shown. The fluid end section **800** comprises the housing **702**, but it also comprises another embodiment of a packing seal system attached to the rear surface **706** of the housing **702**. In contrast to the fluid end

section **700**, the fluid end section **800** does not comprise a stuffing box **198** attached to the rear surface **706** of the housing **702**. Rather, the rear retainer **762** is attached directly to the rear surface **706** of the housing **702** using a plurality of second fasteners **802**. The second fasteners **802** are identical to the second fasteners **196**, but the second fasteners **802** may be shorter in length. [0329] Continuing with FIGS. **111-113**, instead of a stuffing box housing and plunger packing, the fluid end section **800** comprises one and only one packing seal **804** installed within the second counterbore **220** formed in the third section **76** adjacent the rear surface **706**. The packing seal **804** is surrounded by a wear ring **806** also installed within the second counterbore **220**. A rear surface **812** of the packing seal **804** engages the second metal ring **306** installed within the rear retainer **762**.

[0330] Continuing with FIG. **113**, the packing seal **804** is annular and has opposed front and rear surfaces **810** and **812** joined by inner and outer intermediate surfaces **814** and **816**. The inner intermediate surface **814** of the packing seal **804** surrounds and engages an outer surface of the plunger **216**, while the outer intermediate surface **814** seals against the wear ring **806**.

[0331] Turning to FIGS. **114-116**, the packing seal **804** comprises an energizing component **817** installed within an elastomeric body **818**. The energizing component **817** is installed within the front surface **810** of the packing seal **804** and is configured to expand radially when compressed longitudinally. Such expansion causes the inner intermediate surface **814** to tightly seal against the outer surface of the plunger **216** and the outer intermediate surface **816** to tightly seal against the walls of the housing **702** or another component installed therein.

[0332] The energizing component **817** shown in FIG. **116** comprises a plurality of stacked metal pieces **820** having a V-shaped cross-section that function as a spring. Specifically, the packing seal **804** is known in the art as a multi-contact V-nested spring seal. In alternative embodiments, the energizing component may comprise other components known in the art that expand radially when compressed longitudinally. In further alternative embodiments, the energizing component may comprise one or more coiled springs configured to expand the seal regardless of any longitudinal compression.

[0333] The inner and outer surfaces **814** and **816** of the seal **804** further comprise a plurality of seal lips **822**. The seal lips **822** help effectuate sealing during operation. As the seal lips **822** wear over time, the energizing component **817** expands, pushing the lips **822** tighter against the plunger **216** and the wear ring **806**. Only one packing seal **804** is necessary because the packing seal **804** comprises the energizing component **817**, which allows the packing seal **804** to expand when compressed and further expand as it wears over time without applying any additional compression.

[0334] Continuing with FIGS. **113-116**, a support element **824** may also be installed within the second counterbore **220** and engage the front surface **810** and energizing component **817** of the packing seal **804**. The support element **824** comprises an annular base **826** joined to an annular protrusion **828**. The base **826** may be tapered and engage a tapered surface **830** of the second counterbore **220**, as shown in FIG. **113**. The protrusion **828** projects into the energizing component **817**, as shown in FIG. **116**. In operation, the protrusion **828** helps keep the energizing component **817** expanded so as to maintain a tight seal against the plunger **216**. The packing seal system used with the fluid end section **800** and other similar embodiments are described in U.S. Pat. No. 11,946,465, authored by Barnett et al., and issued on Mar. 13, 2024, the entire contents of which are incorporated herein by reference.

Fluid End Section **900**

[0335] Turning to FIGS. **117-124**, another embodiment of a fluid end section **900** is shown. The fluid end section **900** makes up the fluid end assembly **901**, shown in FIG. **124**. The fluid end section **900** is similar to the fluid end section **800**, with a few exceptions. The fluid end section **900** comprises a housing **902** having opposed front and rear surfaces **904** and **906** joined by an outer intermediate surface **908** and a horizontal bore **910** formed therein. The housing **902** is similar to the housing **702**, shown in FIGS. **95** and **96**, but the housing **902** comprises another embodiment of

a third section **912**.

[0336] Continuing with FIGS. **117-119**, the third section **912** comprises a front portion **914** joined to a rear portion **916**. The front portion **914** has a rectangular cross-sectional shape and is the same height and width as the integral section **712**. A plurality of passages **918** are formed in the front portion **914** that align with a plurality of passages **718** formed in the integral section **712**. The aligned passages **918** and **718** are configured to receive the stay rods **58**. The integral section **712** and the third section **912** are held together by the stay rods **58**. Thus, in contrast to the housing **702**, the integral section **712** and the third section **912** do not comprise any passages for receiving the first fasteners **78**.

[0337] Turning to FIGS. **120-121**, the fluid end section **900** uses two dowel sleeves **93** in each pair of aligned passages **918** and **718**. One dowel sleeve **93** spans between the integral section **712** and the third section **912**. Another dowel sleeve **93** spans between the third section **912** and the sleeve **94** disposed around the stay rod **58**.

[0338] Turning to FIGS. **122** and **123**, the front portion **914** of the third section **912** includes the rear surface **906** of the housing **902**. Like the fluid end section **800**, the rear retainer **762** is attached directly to the rear surface **906** of the housing **902** and holds the one and only one packing seal **804** within a counterbore **920** formed in the third section **912**. A support element **922** is shown installed within the packing seal **804**. The support element **922** is generally identical to the support element **824**, but it includes a cylindrical base **924**. The base **924** abuts an annular surface **926** of the counterbore **920**, as shown in FIG. **123**.

[0339] Continuing with FIGS. **122** and **124**, alternative embodiments of a discharge conduit **930** and a suction conduit **932** are shown attached to the housing **902**. Each discharge conduit **930** has a lower opening **934** joined to a horizontal passage **936**. A plurality of horizontally positioned conduits **938** interconnect each discharge conduit **930** to form a discharge manifold **942**, as shown in FIG. **124**. Upper and lower discharge bores **944** and **946** each comprise a counterbore **948** configured to receive the discharge fitting adapter **504**. The adapter **504** spans between a discharge bore **944** or **946** and a discharge conduit **930**.

[0340] Continuing with FIGS. **122** and **124**, upper and lower intake bores **962** and **964** do not include any counterbores or threaded surfaces for receiving the suction conduit **932**. Instead, the bores **962** and **964** have uniform diameters. The suction conduit **932** is attached directly to the intermediate surface **908** of the housing **902** aligning with a corresponding one of the bores **962** and **964** using a plurality of fasteners (not shown). Each suction conduit **932** extends vertically between the fluid end section **900** and a suction manifold **968**. Each suction conduit **932** has a curved shape that curves away from the discharge manifold **942** so as to provide clearance for the suction manifold **968**.

Fluid End Section **1000**

[0341] Turning to FIGS. **125-142**, another embodiment of a fluid end section **1000** is shown. The fluid end section **1000** is similar to the fluid end section **56** shown in FIGS. **8** and **9**, with a few exceptions. The fluid end section **1000** comprises a housing **1002** having opposed front and rear surfaces **1004** and **1006** joined by an outer intermediate surface **1008** and a horizontal bore **1010** formed therein, as shown in FIG. **127**. The housing **1002** is similar to the housing **60**, shown in FIGS. **9-11**, but the housing **1002** comprises another embodiment of a third section **1012**. The housing **1002** is configured to receive the same inner components as the housing **60**.

[0342] With reference to FIGS. **128-131**, the third section **1012** comprises a front surface **1014** joined to the rear surface **1006** of the housing **1002** by a portion of the intermediate surface **1008** and a portion of the horizontal bore **1010**. Like the third section **76**, the outer intermediate surface **1008** of the third section **1012** varies in diameter such that the third section **1012** comprises a front portion **1016** joined to a rear portion **1018**. The front and rear portions **1016** and **1018** may also be referred to as a front mounting flange **1016** and a rear mounting flange **1018**. A medial portion **1020** of the third section **1012** is positioned between the front and rear portion **1016** and **1018** and

has lesser diameter than that of the front and rear portions **1016** and **1018**.

[0343] The front portion **1016** of the third section **1012** is identical to the front portion **180** of the third section **76** shown in FIGS. **25** and **26**. The front portion **1016** comprises the plurality of passages **184** for receiving the fasteners **78** and the fasteners **78** extend over the medial portion **1020** of the third section **1016**, as shown in FIG. **127**. The nut **230** is then installed on the second end **228** of each first fastener **78**.

[0344] The rear portion **1018** of the third section **1012** is different from that of the third section **76** shown in FIGS. **25** and **26**. The rear portion **1018** comprises a plurality of passages **1022** that interconnect a front surface **1024** of the rear portion **1018** and the rear surface **1006** of the housing **1002**, as shown in FIG. **131**.

[0345] Instead of threading second fasteners into threaded openings formed in the third section, a plurality of second fasteners **1026** each extend all of the way through the corresponding passage **1022** and a first threaded end **1028** of each second fastener **1026** is positioned over the medial portion **1020** of the third section **1012**, as shown in FIG. **127**. A blind nut **1030** is installed on each first threaded end **1028** and is turned until it is tightly secured on the corresponding end **1028**.

[0346] Following installation of the second fasteners **1026** into the passages **1022** and assembling of the blind nuts **1030**, the retention plate **206**, stuffing box **198**, and rear retainer **200** are installed on the second fasteners **1026** in the same manner as the fluid end section **56**. The components are tightly secured together by installing a washer **1059** and a nut **1060** on a second threaded end **1062** of each second fastener **1026**, like it is done on the fluid end section **56**. The plunger packing **300**, packing nut **290**, and the plunger **216** are also installed within the fluid end section **1000** in the same manner as the fluid end section **56**.

[0347] As shown in FIGS. **125-127**, when the fluid end section **1000** is assembled, the second ends **228** of the first fasteners **78** and the corresponding nuts **230** and the first ends **1028** of the second fasteners **1026** and the corresponding nuts **1030** are in a spaced-relationship and face each other. Both ends **228** and **1026** and corresponding nuts **230** and **1030** are positioned between the front and rear portions **1016** and **1018** and both ends and corresponding nuts overlap the medial portion **1020** of the third section **1012**. When assembled, the second fasteners **1026** extend along an axis that is parallel to a longitudinal axis **1032** of the housing **1002**, as shown in FIG. **127**.

[0348] The rear portion **1018** of the third section **1012** also comprises the dowel openings **207** configured to receive third alignment dowels **242** used to align the retention plate **206** on the third section **1012**, as shown in FIGS. **129** and **142**. The rear portion **1018** further comprises the threaded openings **202** configured to receive the third fasteners **204** used to secure the retention plate **206** to the third section **1012** of the housing **1002**, as also shown in FIGS. **129** and **142**.

[0349] Turning back to FIG. **125**, a reaction washer **1034** is also installed on the blind nut **1030** during assembly of the fluid end section **1000**. The reaction washer **1034** provides a mechanism for resisting torque applied to the blind nuts **1030** when other parts of the fluid end section **1000** are assembled. The blind nut **1030** is a flanged nut and comprises a flanged section **1036** joined to a drive section **1038**, as shown in FIGS. **132-135**. The flanged section **1036** functions as a washer. The drive section **1038** comprises a standard 12-point drive, sometimes called a double hex drive. A series of grooves **1040** are formed in the outer surface of the drive section **1038** and extend radially around the periphery of the outer surface of the drive section **1038**. The grooves **1040** are configured to receive a seal **1042**, such as an O-ring seal. The seal **1042** is configured to hold the reaction washer **1034** on the blind nut **1030**, as shown in FIGS. **138** and **139**. The blind nut **1030** further comprises a threaded blind bore **1044** having a base **1046**. An opening **1048** may be formed in the base **1046** for viewing the interior of the blind bore **1044**.

[0350] With reference to FIGS. **136** and **137**, the reaction washer **1034** comprises a pair of torque reaction arms **1054** positioned on opposite sides of a central opening **1052**. The reaction arms **1054** are joined by a concave surface **1056** sized to conform to the outer cylindrical surface of the medial portion **1020** of the third section **1012**, as shown in FIG. **138**. The central opening **1052** comprises

a 12-point wall **1058** that is congruent to the 12-point drive section **1038** of the blind nut **1030**. [0351] The reaction washer **1034** is shaped to slide over the drive section **1038** of the blind nut **1030** and is held on the drive section **1038** by the seal **1042**, as shown in FIG. **139**. When installed thereon, at least one of the reaction arms **1054** of the reaction washer **1034** engages the medial portion **1020** of the third section **1012**. When turning a nut **1060** on a second end **1062** of each second fastener **1026**, the reaction washers **1034** provide a mechanism for resisting the torque applied to the blind nuts **1030**, allowing the nuts **1060** to be tightened or loosened without the blind nuts **1030** spinning freely. Specifically, rotation of the blind nut **1030** is prevented by the engagement of one of the reaction arms **1054** with the medial portion **1020** of the housing **1002**, as shown in FIG. **138**. After assembly, the reaction washers **1034** may be removed from the blind nuts **1030**. Alternatively, the reaction washers **1034** may be left in place so as to prevent rotation of the blind nuts **1030** during operation, thereby reducing the likelihood of the second fasteners **1026** from coming loose.

[0352] Turning back to FIGS. **127** and **130**, the third section **1012** of the housing **1002** comprises the counterbore **208** for receiving the hardened insert **210**. The insert **210** may also be referred to as a wear ring **210**. In contrast to the third section **76** shown in FIGS. **9** and **14**, a seal groove **1064** is also formed within the third section **1012** at a corner of the counterbore **208**. The seal groove **1064** is configured to receive a seal **1066**, as shown in FIG. **140**. The seal **1066** shown in the figures is an O-ring. In alternative embodiments, the seal **1066** may comprise other types of seals known in the art. During operation, the seal prevents fluid from leaking around the hardened insert **210** when the insert **210** is installed within the counterbore **208**.

[0353] Similarly, a seal groove **1068** is formed in the corner of the second counterbore **220** formed in the third section **1012**. The seal groove **1068** is configured to receive a seal **1070**, such as an O-ring shown in FIG. **141**. During operation, the seal **1070** prevents fluid from leaking around the wear ring **222**.

Fluid End Section **1100**

[0354] Turning to FIGS. **143-168**, another embodiment of a fluid end section **1100** is shown. The fluid end section **1100** is similar to the fluid end section **56** shown in the FIGS. **7-9**, with a few exceptions. The fluid end section **1100** comprises a housing **1102** having opposed front and rear surfaces **1104** and **1106** joined by an outer intermediate surface **1108** and a horizontal bore **1110** formed therein, as shown in FIG. **145**. The housing **1102** has the same general shape as the housing **60**, but the housing **1102** comprises two sections instead of three. The housing **1102** comprises a first section **1112** attached to a second section **1114** using a plurality of fasteners **1116**, as shown in FIG. **146**. The walls of the horizontal bore **1110** are shaped like those within the housing **60** and are configured to receive the same inner components as the housing **60**, as shown in FIGS. **9** and **145**.

[0355] The first section **1112** is identical to the first section **72** shown in FIGS. **17** and **18**, but the first section **1112** is extended so as to also include the suction bores **160** and **162**, as shown in FIG. **145**. Thus, the middle or second section **74** making up the housing **60** has been removed. The first section **1112** comprises the plurality of passages **86** and corresponding notches **82** configured to receive a plurality of stay rods **58** in a one-to-one relationship, as shown in FIGS. **152-155**. The first section **1112** further comprises the plurality of threaded openings **96** formed in its rear surface **1118** for receiving a first threaded end **1120** of each of the fasteners **1116**, as shown in FIGS. **151** and **153**.

[0356] The second section **1114** of the housing **1102** is similar in shape to the third section **76** of the housing **60** shown in FIGS. **25** and **26**. The second section **1114** comprises a front surface **1122** joined to the rear surface **1106** by a portion of the outer intermediate surface **1108** and a portion of the horizontal bore **1110**, as shown in FIG. **151**. The outer intermediate surface **1108** comprises a front mounting flange **1124** positioned adjacent the front surface **1122** and a rear mounting flange **1126** positioned adjacent the rear surface **1106**, as shown in FIGS. **156** and **157**.

[0357] The front and rear mounting flanges **1124** and **1126** break up the intermediate surface **1108**

of the second section **1114** such that a front projecting portion **1128** of the intermediate surface **1108** exists between the front surface **1122** and the front mounting flange **1124**, and a rear projecting portion **1130** of the intermediate surface **1108** exists between the rear surface **1106** and the rear mounting flange **1126**, as shown in FIGS. **156** and **157**. Finally, a medial portion **1132** of the intermediate surface **1108** exists between the front and rear mounting flanges **1124** and **1126**. The front and rear mounting flanges **1124** and **1126** each have a greater outer diameter than that of the front, rear, and medial portions **1128**, **1130**, and **1132** of the intermediate surface **1108**, as shown in FIGS. **150** and **151**. The front mounting flange **1124** further has a greater outer diameter than the rear mounting flange **1126**.

[0358] Continuing with FIGS. **156** and **157**, a plurality of first passages **1134** are formed within the front mounting flange **1124**. The first passages **1134** surround the horizontal bore **1110**. Each first passage **1134** is configured to receive one of the first fasteners **1116**, as shown in FIG. **151**. When the first and second sections **1112** and **1114** are joined together, a space exists around the front portion **1128** and between the rear surface **1118** of the first section **1112** and the front mounting flange **1124**. As will be described herein, the space is sized to receive the front spacer sleeve **1136**, as shown in FIG. **150**. A plurality of clearance notches **1138** are also formed in the outer intermediate surface **1108** of the front mounting flange **1124**, as shown in FIG. **156**. Each clearance notch **1138** is sized to partially surround a corresponding sleeve **94** and stay rod **58** when the fluid end section **1100** is assembled.

[0359] Continuing with FIGS. **156** and **157**, a plurality of second passages **1140** are formed within the rear mounting flange **1126**. As will be described herein, the second passages **1140** are configured to receive a plurality of second fasteners **1026** in a one-to-one relationship, as shown in FIG. **147**. When a stuffing box **198** is attached to the second section **1114**, a space exists around the rear portion **1130** of the second section **1114** and between the rear surface **1106** and the rear mounting flange **1126**. As will be described herein, the space is sized to receive a rear spacer sleeve **1144**, as shown in FIGS. **145-147**.

[0360] A plurality of dowel openings **1146** are also formed with the rear mounting flange **1126**, each opening facing the rear surface **1106** of the housing **1102** and configured to receive a portion of a first dowel pin **1148**, as shown in FIG. **151**. Each first dowel pin **1148** is configured to extend between the rear mounting flange **1126** and the rear spacer sleeve **1144**. A plurality of threaded openings **1150** are further formed in the rear mounting flange **1126**, each opening facing the rear surface **1106** of the housing **1102**. The threaded openings **1150** are preferably diametrically opposed to one another and are each configured to receive one of a plurality of third fasteners **1152**, as shown in FIG. **146**. The third fasteners **1152** are configured to secure the rear spacer sleeve **1144** to the second section **1114** of the housing **1102**.

[0361] Continuing with FIGS. **150** and **151**, like the third section **76** shown in FIG. **9**, the second section **1114** comprises a counterbore **1115** for receiving a hardened insert **1123** and a counterbore **1117** for receiving a wear ring **1125**, as shown in more detail in FIGS. **164-166**. The counterbores **1115** and **1117** are like the counterbores **208** and **220** shown in FIGS. **140** and **141** but a side wall **1119** and **1121** of each corresponding counterbore **1115** and **1117** is tapered. The insert **1123** and wear ring **1125** are like the insert **210** and the wear ring **222** shown in FIGS. **164-166**, but the insert **1123** and the wear ring **1125** each have a tapered outer surface to correspond with the taper of the corresponding counterbore **1115** or **1117**. The corresponding tapers help secure or taper lock the components into the corresponding counterbore **1115** and **1117**.

[0362] The second section **1114** further comprises the seal grooves **1064** and **1068**, shown in FIGS. **164-166**. However, the seal grooves **1064** and **1068** are formed in the center of a side wall **1119** and **1121** of each corresponding counterbore **1115** and **1117**, instead of being formed in the corner of each counterbore **1115** and **1117**. The seals **1066** and **1070** are installed within the corresponding seal grooves **1064** and **1068** and are configured to prevent fluid from leaking around the insert **1123** and the wear ring **1125**.

[0363] Continuing with FIG. 165, a stress relief cutout 275 may be further formed in the outer surface 336 of the fluid routing plug 132 adjacent the annular shoulder 400. The annular shoulder 400 does not contact the insert 1123 in this area. The absence of contact reduces the stress on the annular shoulder 400 during operation, helping to extend the life of the fluid routing plug 132.

[0364] Turning back to FIGS. 148-151, the housing 1102 is also different from the housing 60 because it comprises the front spacer sleeve 1136. The front spacer sleeve 1136 comprises opposed front and rear surfaces 1172 and 1174 joined by a central opening 1176 and an outer intermediate surface 1178, as shown in FIGS. 158 and 159. A plurality of passages 1180 are formed in the front spacer sleeve 1136 surrounding the central opening 1176. The plurality of passages 1180 are each sized to receive one of the first fasteners 1116 and are positioned to correspond with the threaded openings 96 formed in the rear surface 1118 of the first section 1112, as shown in FIG. 151.

[0365] A plurality of clearance notches 1182 are formed in the outer intermediate surface 1178 of the front spacer sleeve 1136, as shown in FIGS. 158 and 159. Each clearance notch 1182 is sized to partially surround a corresponding sleeve 94 and stay rod 58 when the fluid end section 1100 is assembled. The clearance notches 1182 align with the clearance notches 1138 formed in the front mounting flange 1124 of the second section 1114 of the housing 1102 when the fluid end section 1100 is assembled, as shown in FIG. 144.

[0366] The central opening 1176 of the front spacer sleeve 1136 is sized to receive the front portion 1128 of the second section 1114 of the housing 1102 such that the front spacer sleeve 1136 surrounds the front portion 1128 of the second section 1114, as shown in FIGS. 150 and 151. When the front portion 1128 is installed within the central opening 1176, an outer diameter of the front spacer sleeve 1136 is equal or substantially equal to and aligned with an outer diameter of the front mounting flange 1124.

[0367] During operation, the central opening 1176 of the front spacer sleeve 1136 is sized so that it never applies any meaningful compressive force to the front portion 1128 of the second section 1114 of the housing 1102. Specifically, the front spacer sleeve 1136 does not constrain the expansion of the front portion 1128 of the second section 1114 when the annular insert 210 is pressed into the first counterbore 208. This results in more uniform deflection of the insert 210 during installation. The stress resulting from such deflection is therefore also more uniform. As a result, stress concentration areas within the insert 210 are eliminated or significantly reduced, thereby increasing the life of the insert and extending the time between maintenance intervals for the fluid end section 1100.

[0368] The front spacer sleeve 1136 is also sized so that the front surface 1172 of the spacer sleeve 1136 shares the compressive load with the front surface 1122 of the second section 1114 of the housing 1102. Sharing of the compressive load between the two components reduces the stress on the front surface 1122 of the second section 1114. The rear surface 1174 of the front spacer sleeve 1136 provides a reactionary surface for the front surface of the front mounting flange 1124. Providing such surface reduces the bending deflection of the front mounting flange 1124 and consequently the bending stress on the front mounting flange 1124. The benefits provided by the front spacer sleeve 1136 increase the life of the corresponding components, thereby extending the time between maintenance intervals for the fluid end section 1100.

[0369] With reference to FIGS. 145-147, prior to assembling the housing 1102, the seal 1066 is installed within the first seal groove 1064 formed in the second section 1114 and the insert 1123 is press-fit or interference fit into the first counterbore 208. The seal 1066 shown in the figures is an O-ring seal, but other types of seals known in the art may also be used. The insert 210 will be described in more detail later herein.

[0370] Following installation of the seal 1066 and the insert 210, the housing 1102 is assembled by threading a first threaded end 1120 of each of the first fasteners 1116 into a corresponding one of the threaded openings 96 formed in the first section 1112, as shown in FIG. 151. Once installed therein, the first fasteners 1116 project from the rear surface 1118 of the first section 1112. The

front spacer sleeve **1136** is then slid onto the first fasteners **1116** using the corresponding passages **1180**. The front spacer sleeve **1136** is aligned such that the clearance notches **1182** correspond with the stay rod passages **86** formed within the first section **1112**, as shown in FIG. **144**.

[0371] After the front spacer sleeve **1136** is in place, the second section **1114** of the housing **1102** is slid onto the first fasteners **1116** using the corresponding passages **1134** formed in the front mounting flange **1124** as shown in FIG. **151**. At the same time, the front portion **1128** of the second section **1122** is inserted into the central opening **1176** of the front spacer sleeve **1136** until the front surface **1122** of the second section **1114** abuts the rear surface **1118** of the first section **1112**.

[0372] When the second section **1114** is brought together with the first section **1112**, a second threaded end **1184** of each first fastener **1116** projects from the front mounting flange **1124** and extends over the medial portion **1132** of the intermediate surface **1108**, as shown in FIGS. **144-147**. At the same time, the front spacer sleeve **1136** is sandwiched between the rear surface **1118** of the first section **1112** and the front mounting flange **1124** such that the front and rear surfaces **1172** and **1174** of the front spacer sleeve **1136** abut the corresponding surfaces. The second section **1114** of the housing **1102** is aligned on the first fasteners **1116** such that the clearance notches **1138** formed in the front mounting flange **1124** align with the clearance notches **1182** formed in the front spacer sleeve **1136**, as shown in FIG. **144**.

[0373] The assembled sections **1112** and **1114** are secured together by installing a threaded nut **1186** onto the second end **1184** of each first fastener **1116**, as shown in FIGS. **144** and **146**. The nut **1186** shown in the figures is a flanged nut having a 12-point drive section. The nut **1186** is turned on the second end **1184** of the fastener **1116** to the desired specification or until it tightly engages the front mounting flange **1124**. When installed thereon, each nut **1186** overlaps the medial portion **1132** and is positioned intermediate the front and rear mounting flanges **1124** and **1126** of the second section **1114** of the housing **1102**.

[0374] Twelve first fasteners **1116** and corresponding threaded openings **96**, passages **1180** and **1134**, and nuts **1186** are shown in the figures. In alternative embodiments, more or less than twelve first fasteners **1116** and corresponding openings **96**, passages **1180** and **1134**, and nuts **1186** may be used to assemble the housing **1102**. When the housing **1102** is assembled, the first fasteners **1116** are each positioned along an axis that is parallel to a longitudinal axis **1188** of the housing **1102**.

[0375] The first fasteners **1116** are identical to the first fasteners **78** shown in FIG. **15** but are shorter in length because each fastener **1116** traverses a shorter distance. The first fastener **1116** shown in the figures is a threaded stud. In alternative embodiments, other types of fasteners known in the art may be used instead of a threaded stud. For example, screws or bolts may be used to secure the sections together. In further alternative embodiments, the nut **1186** may comprise the three-piece nut **90**, shown in FIG. **19**, or other types of threaded nuts known in the art.

[0376] To remove a section **1112** or **1114** or remove the front spacer sleeve **1136**, the nut **1186** is unthreaded from the second end **1184** of each first fastener **1116**. The sections **1112** and **1114** and the rear spacer sleeve **1144** may then be pulled apart, as needed. If the first section **1112** is being replaced, the first fasteners **1116** are also unthreaded from the threaded openings **96**. The components installed within the housing **1102** may also be removed, as needed, prior to disassembling the housing **1102**.

[0377] With reference to FIG. **147**, like the rear portion **1018** of the housing **1002** shown in FIG. **131**, the rear mounting flange **1126** is configured to have the second fasteners **1026** extend entirely through the rear mounting flange **1126**. Like the fluid end section **1000**, the blind nut **1030** and reaction washer **1034** are installed on the first threaded end **1028** of the second fastener **1026**. The stuffing box **198** and the rear retainer **200** are then installed on the second fasteners **1026** in the same manner as the fluid end section **1000**.

[0378] Continuing with FIGS. **145-147**, in contrast to the fluid end section **1000**, the fluid end section **1100** uses the rear spacer sleeve **1144** instead of the retention plate **206**. The washer **1059** and the nut **1060** are installed on the second threaded end **1062** of each second fastener **1026** to

secure the components together. The plunger packing **300**, the packing nut **290**, and the plunger **216** are installed within the fluid end section **1100** in the same manner as the fluid end section **1000**.

[0379] With reference to FIGS. **162** and **163**, the rear spacer sleeve **1144** comprises opposed front and rear surfaces **1190** and **1192** joined by a central opening **1194** and an outer intermediate surface **1196**. A plurality of passages **1198** are formed in the rear spacer sleeve **1144** surrounding the central opening **1194**. The plurality of passages **1198** are each sized to receive one of the second fasteners **1026**, as shown in FIG. **147**. A plurality of counterbored passages **1200** are also formed within the rear spacer sleeve **1144**. The counterbored passages **1200** are positioned to align with the threaded openings **1150** formed in the rear mounting flange **1126** and are configured to receive a corresponding third fastener **1152**, as shown in FIG. **146**. A counterbore **1202** of each counterbored passage **1200** opens on the rear surface **1192** of the rear spacer sleeve **1144**. Finally, a plurality of dowel passages **1204** are formed within the rear spacer sleeve **1144**. Each dowel passage **1204** is positioned to align with a corresponding dowel opening **1146** formed in the rear mounting flange **1126**, as shown in FIG. **151**.

[0380] The central opening **1194** of the rear spacer sleeve **1144** is sized to receive the rear portion **1130** of the second section **1114** of the housing **1102** such that the rear spacer sleeve **1144** surrounds the rear portion **1130** of the second section **1114**, as shown in FIGS. **145-147**. When the rear portion **1130** is installed within the central opening **1194**, an outer diameter of the rear spacer sleeve **1144** is equal to or substantially equal to and aligned with an outer diameter of the rear mounting flange **1126**.

[0381] Like the front spacer sleeve **1136**, the central opening **1194** of the rear spacer sleeve **1144** is sized so that it never applies any meaningful compressive force to the rear portion **1130** of the second section **1114** of the housing **1102**. Specifically, the rear spacer sleeve **1144** does not constrain the expansion of the rear portion **1130** of the second section **1114** when the wear ring **222** is pressed into the second counterbore **220**. This results in more uniform deflection of the wear ring **222** during installation. The stress resulting from such deflection is therefore also more uniform. As a result, stress concentration areas within the wear ring **222** are eliminated or significantly reduced, thereby increasing the life of the wear ring **222** and extending the time between maintenance intervals for the fluid end section **1100**.

[0382] Like the front spacer sleeve **1136**, the rear spacer sleeve **1144** is also sized so that the rear surface **1192** of the spacer sleeve **1144** shares the compressive load with the rear surface **1106** of the housing **1102**. Sharing of the compressive load between the two components reduces the stress on the rear surface **1106** of the housing **1102**. The front surface **1190** of the rear spacer sleeve **1144** provides a reactionary surface for the rear surface of the rear mounting flange **1126**. Providing such surface reduces the bending deflection of the rear mounting flange **1126** and consequently the bending stress on the rear mounting flange **1126**. The benefits provided by the rear spacer sleeve **1144** increase the life of the corresponding components, thereby extending the time between maintenance intervals for the fluid end section **1100**.

[0383] Continuing with FIGS. **145-147**, to attach the rear spacer sleeve **1144** to the housing **1102**, a first dowel pin **1148** is inserted into a corresponding one of the dowel openings **1146** formed on the rear mounting flange **1126** of the second section **1114**, as shown in FIG. **151**. The rear spacer sleeve **1144** is then brought together with the second section **1114** of the housing **1102** such that the first dowel pins **1148** are inserted into the dowel passages **1204**, the second fasteners **1026** are disposed within the passages **1140**, and the threaded openings **1150** are aligned with the counterbored passages **1200**, as shown in FIGS. **146** and **147**. When in such position, the front surface **1190** of the rear spacer sleeve **1144** abuts the rear surface **1106** of the housing **1102** and the rear portion **1130** of the second section **1114** is installed within the central opening **1194** of the rear spacer sleeve **1144**.

[0384] The rear spacer sleeve **1144** is secured to the rear surface **1106** of the housing **1102** by

installing a third fastener **1152** within a corresponding one of the aligned threaded openings **1150** and counterbored passages **1200**, as shown in FIG. **146**. The third fasteners **1152** shown in the figures are screws. The third fasteners **1152** are torqued within the threaded openings **1150** until a head **1206** of each fastener **1152** abuts a base **1208** of each counterbored passage **1200**, thereby rigidly securing the rear spacer sleeve **1144** to the housing **1102**. When the rear spacer sleeve **1144** is attached to the housing **1102**, the rear surface **1106** of the rear spacer sleeve **1144** is positioned flush with the rear surface **1106** of the housing **1102**.

[0385] As shown in FIGS. **143** and **144**, when the fluid end section **1100** is assembled, the second ends **1184** of the first fasteners **1116** and the corresponding nuts **1186** and the first ends **1028** of the second fasteners **1026** and the corresponding nuts **1030** are in a spaced-relationship and face each other. Both ends **1184** and **1028** and corresponding nuts **1186** and **1030** are positioned between the first and rear mounting flanges **1124** and **1126** and both ends and corresponding nuts overlap the medial portion **1132** of the second section **1114**. When assembled, the second fasteners **1026** extend along an axis that is parallel to the longitudinal axis **1188** of the housing **1102**, as shown in FIG. **147**.

Fluid End Section **1300**

[0386] Turning to FIGS. **169-174**, another embodiment of a fluid end section **1300** is shown. The fluid end section **1300** is similar to the fluid end section **1100** shown in FIGS. **143** and **144** with a few exceptions. The fluid end section **1300** comprises a housing **1302** having opposed front and rear surfaces **1304** and **1306** joined by an outer intermediate surface **1308** and a horizontal bore **1310** formed therein, as shown in FIG. **171**. The housing **1302** has the same general shape as the housing **1102**, but the housing **1302** is of single-piece construction. The housing **1302** does not include multiple sections joined together by fasteners.

[0387] The walls of the horizontal bore **1310** are shaped identical to those within the housing **1102** and are configured to receive the same inner components as the housing **1102**, as shown in FIGS. **145** and **172**. However, the housing **1302** is shown with the counterbores **208** and **220** and corresponding inserts **210** and wear ring **222** installed therein. In alternative embodiments, the counterbores **1115** and **1117** may be formed within the housing **1302** and have the corresponding insert **1123** and wear ring **1125** installed therein.

[0388] The outer intermediate surface **1308** of the housing **1302** comprises a first section **1312** integrally joined to a second section **1314**. The first section **1312** is shaped identically to the first section **1112** and comprises the plurality of passages **86** and corresponding notches **82** configured to receive a plurality of stay rods **58** in a one-to-one relationship, as shown in FIGS. **169** and **170**. The second section **1314** of the housing **1302** comprises a cylindrical portion **1316** that transitions into the rear mounting flange **1126** and the rear projecting portion **1130** of the housing **1302**, as shown in FIGS. **171-174**. Because the housing **1302** is a single piece, the housing **1302** does not include the front spacer sleeve **1136**.

[0389] The cylindrical portion **1316** of the second section **1314** of the housing **1302** is configured to house the same components as the second section **1114** of the housing **1102**, shown in FIGS. **145** and **172**. The rear mounting flange **1126** and rear portion **1130** of the housing **1302** are configured to mate with the rear spacer sleeve **1144**, the stuffing box **198**, and the rear retainer **200** in the same manner as the fluid end section **1100**, shown in FIG. **145**.

[0390] Continuing with FIG. **171**, the second fasteners **1026** extend through the rear mounting flange **1126**, rear spacer sleeve **1144**, stuffing box **198**, and rear retainer **200** in the same manner as the fluid end section **1100**. The second fasteners **1026** are secured on opposite ends **1028** and **1062** using the blind nuts **1030** and the nuts **1060** and corresponding washers **1059**. While not shown, a reaction washer **1034** and seal **1042** may also be installed on each of the blind nuts **1030**. The plunger packing **300**, the packing nut **290**, and the plunger **216** are installed within the fluid end section **1300** in the same manner as the fluid end section **1100**, as shown in FIG. **172**. When the fluid end section **1300** is assembled, the ends **1028** of the second fasteners **1026** and the blind nuts

1030 are in a spaced-relationship with and face the cylindrical portion **1316** of the second section **1314** of the housing **1302**, as shown in FIGS. **169-172**.

Stress Relief Cutout

[0391] Referring now to FIGS. **175** and **176**, stress relief cutout **275** of the fluid routing plug **132** is shown in greater detail than previously depicted in FIGS. **145**, **164**, and **165**. Stress relief cutout **275** comprises a first radius **533**, a linear section **534**, and a second radius **535**. The first radius **533** is a convex radius tangent to the annular shoulder **400**, transitioning from the annular shoulder **400** to the linear section **534**. The second radius **535** is a concave radius that is tangent to both the outer intermediate surface **336** and the linear section **534**, smoothly transitioning between the two. As mentioned above, the stress relief cutout **275** does not contact the hardened insert **1123**, or hardened insert **210** if the non-tapered embodiment is used.

[0392] While not discernible at the scale shown in the previous figures, FIG. **175** reveals that the first bevel **426** of the hardened insert **1123** comprises multiple sections, including a first radius **536**, a linear section **537**, and a second radius **538**. The first radius **536** is convex and tangent to both the inner intermediate surface **420** and the linear section **537**. The second radius **538** is also convex, larger than the first radius **536**, and tangent to both the linear section **537** and the front surface **416**. In other embodiments, the first bevel **426** may consist of only the linear section **537**, or the linear section **537** and a single radius, either **536** or **538**. Additionally, in other embodiments the first radius **536** may be the same size as, or larger than, the second radius **538**. It should be understood that “linear” in the context of portions of the ring-shaped insert, refers to the linear nature of the cross-section of the linear section **537** insert **1123**, rather than the surface itself.

[0393] The hardened insert **1123** contacts the annular shoulder **400** at the point where the first radius **533** joins the annular shoulder **400**, as shown in FIG. **175**. This point is defined as the minimum radial contact point **P1**. The minimum radial contact point **P1** is at a radial distance **D1** from the section of the outer intermediate surface **336** of the fluid routing plug **132** that contains the second annular groove **392**. The second annular groove **392** is shown in FIG. **165**. The minimum radial contact point **P1** represents a circular line of contact that extends around the front surface **416** of hardened insert **1123**.

[0394] In operation, the force generated by fluid pressure is distributed across the contacting surfaces of the annular shoulder **400** and the front surface **416** of the hardened insert **1123**. The radial distance **D1** locates the minimum radial contact point **P1** on the front surface **416** of the hardened insert **1123**, positioned radially outward from the inner intermediate surface **420**, toward the outer intermediate surface **422**, and away from any edge modifications, such as a first bevel **426**. Increasing the radial distance **D1** reduces or eliminates chipping or spalling of the front surface **416**. Additionally, the second radius **535** reduces the stress concentration at the transition between the linear section **534** and the section of outer intermediate surface **336** that contains the second annular groove **392**.

[0395] Another embodiment of a stress relief cutout **175** is shown in FIGS. **177** and **178**. Stress relief cutout **175** comprises a first radius **539** and a linear section **540**. The first radius **539** is a concave radius that is tangent to both the outer intermediate surface **336** and the linear section **540**, smoothly transitioning between the two. The linear section **540** is tangent to the first radius **539** and creates an angle $\beta 2$ with the annular surface **400** as shown in FIG. **178**. In this embodiment, the angle $\beta 2$ is 5 degrees but may be any angle between 0 and 90 degrees, exclusive of 0 degrees. The stress relief cutout **175** does not contact the hardened insert **1123**, or hardened insert **210** if the non-tapered embodiment is used.

[0396] The hardened insert **1123** contacts the annular shoulder **400** at the point where the linear section **540** joins the annular shoulder **400**, as shown in FIG. **177**. This point is defined as the minimum radial contact point **P2**. The minimum radial contact point **P2** is at a radial distance **D2** from the section of the outer intermediate surface **336** of the fluid routing plug **132** that contains the second annular groove **392**. The second annular groove **392** is shown in FIG. **165**. The minimum

radial contact point P2 represents a circular line of contact that extends around the front surface **416** of hardened insert **1123**.

[0397] In operation, the radial distance D2 locates the minimum radial contact point P2 on the front surface **416** of the hardened insert **1123**, positioned radially outward from the inner intermediate surface **420**, toward the outer intermediate surface **422**, and away from any edge modifications, such as a first bevel **426**. This reduces or eliminates chipping or spalling of the hardened insert **1123**. Additionally, the first radius **539** reduces the stress concentration at the transition between the linear section **540** and the section of outer intermediate surface **336** that contains the second annular groove **392**. Further, the shallow transition angle $\beta 2$ reduces the stress concentration in the fluid routing plug **132** created by the transition from the annular surface **400** to the stress relief cutout **175**.

[0398] Another embodiment of a stress relief cutout **375** is shown in FIGS. **179** and **180**. The stress relief cutout **375** comprises a first radius **541** and a second radius **542**. The first radius **541** is a convex radius that is tangent to both the annular shoulder **400** and the second radius **542**, transitioning from the annular shoulder **400** to the second radius **542**. The second radius **542** is a concave radius that is tangent to both the outer intermediate surface **336** and the first radius **541**, smoothly transitioning between the two. The stress relief cutout **375** does not contact the hardened insert **1123**, or hardened insert **210** if the non-tapered embodiment is used.

[0399] The hardened insert **1123** contacts the annular shoulder **400** at the point where the first radius **541** joins the annular shoulder **400**, as shown in FIG. **179**. This point is defined as the minimum radial contact point P3. The minimum radial contact point P3 is at a radial distance D3 from the section of the outer intermediate surface **336** of the fluid routing plug **132** that contains the second annular groove **392**. The second annular groove **392** is shown in FIG. **165**. The minimum radial contact point P3 represents a circular line of contact that extends around the front surface **416** of hardened insert **1123**.

[0400] In operation, the radial distance D3 locates the minimum radial contact point P3 on the front surface **416** of the hardened insert **1123**, positioned radially outward from the inner intermediate surface **420**, toward the outer intermediate surface **422**, and away from any edge modifications, such as a first bevel **426**. This reduces or eliminates chipping or spalling of the hardened insert **1123**. Additionally, the second radius **542** reduces the stress concentration at the transition between the first radius **541** and the section of outer intermediate surface **336** that contains the second annular groove **392**. Further, the first radius **541** reduces the stress concentration in the fluid routing plug **132** created by the transition from the annular surface **400** to the stress relief cutout **375**.

[0401] Another embodiment of a stress relief cutout **475** is shown in FIGS. **181** and **182**. The stress relief cutout **475** comprises a first radius **543**, a second radius **544**, and a third radius **545**. The first radius **543** is a convex radius that is tangent to both the annular shoulder **400** and the second radius **544**, transitioning from the annular shoulder **400** to the second radius **544**. The second radius **544** is a concave radius that is tangent to both the first radius **543** and the third radius **545**, smoothly transitioning between the two. The third radius **545** is a convex radius that is tangent to both the second radius **544** and the outer intermediate surface **336**. The stress relief cutout **475** does not contact the hardened insert **1123**, or hardened insert **210** if the non-tapered embodiment is used.

[0402] The hardened insert **1123** contacts the annular shoulder **400** at the point where the first radius **543** joins the annular shoulder **400**, as shown in FIG. **181**. This point is defined as the minimum radial contact point P4. The minimum radial contact point P4 is at a radial distance D4 from the section of the outer intermediate surface **336** of the fluid routing plug **132** that contains the second annular groove **392**. The second annular groove **392** is shown in FIG. **165**. The minimum radial contact point P4 represents a circular line of contact that extends around the front surface **416** of hardened insert **1123**.

[0403] In operation, the radial distance D4 locates the minimum radial contact point P4 on the front surface **416** of the hardened insert **1123**, positioned radially outward from the inner intermediate

surface **420**, toward the outer intermediate surface **422**, and away from any edge modifications, such as a first bevel **426**. This reduces or eliminates chipping or spalling of the hardened insert **1123**. Additionally, the second radius **544** reduces the stress concentration at the transition between the first radius **543** and the third radius **545**. Further, the first radius **543** reduces the stress concentration in the fluid routing plug **132** created by the transition from the annular surface **400** to the stress relief cutout **475** and the third radius **545** reduces the stress concentration in the fluid routing plug **132** created by the transition from the section of outer intermediate surface **336** that contains the second annular groove **392** to the stress relief cutout **475**.

[0404] Another embodiment of a stress relief cutout **575** is shown in FIGS. **183** and **184**. The stress relief cutout **575** comprises a first linear section **546**, a second linear section **547**, a first radius **548**, a third linear section **549**, and a second radius **550**. The first linear section **546** is perpendicular to the annular surface **400** and connects the annular surface **400** to the second linear section **547**. The second linear section **547** is parallel to the annular surface **400**, thus perpendicular to the first linear section **546**, and tangent to the first radius **548** connecting the first linear section **546** to the first radius **548**. The first radius **548** is a concave radius that is tangent to both the second linear section **547** and the third linear section **549**. The third linear section **549** forms an angle β_5 with the second linear section, is tangent to both the first and second radii **548** and **550**, and connects the first radius **548** to the second radius **550**. The second radius **550** is a concave radius that is tangent to both the third linear section **549** and the outer intermediate surface **336**. The stress relief cutout **575** does not contact the hardened insert **1123**, or hardened insert **210** if the non-tapered embodiment is used. In this embodiment, the angle β_5 is 45 degrees but may vary from 0 to 90 degrees, exclusive of both.

[0405] The hardened insert **1123** contacts the annular shoulder **400** at the point where the first linear section **546** joins the annular shoulder **400**, as shown in FIG. **183**. This point is defined as the minimum radial contact point **P5**. The minimum radial contact point **P5** is at a radial distance **D5** from the section of the outer intermediate surface **336** of the fluid routing plug **132** that contains the second annular groove **392**. The second annular groove **392** is shown in FIG. **165**. The minimum radial contact point **P5** represents a circular line of contact that extends around the front surface **416** of hardened insert **1123**.

[0406] In operation, the radial distance **D5** locates the minimum radial contact point **P5** on the front surface **416** of the hardened insert **1123**, positioned radially outward from the inner intermediate surface **420**, toward the outer intermediate surface **422**, and away from any edge modifications, such as a first bevel **426**. This reduces or eliminates chipping or spalling of the hardened insert **1123**. Additionally, the first radius **548**, third linear section **549**, and second radius **550** combine to reduce the stress concentration at the transition between the second linear section **547** and the outer intermediate surface **336**.

[0407] The key feature in all the disclosed embodiments of the stress relief cutouts **175**, **275**, **375**, **475**, and **575** is the location of the minimum radial contact points **P1**, **P2**, **P3**, **P4**, and **P5** radially outward from the inner intermediate surface **420**, toward the outer intermediate surface **422**, and away from any edge modifications, such as a first bevel **426**. Though the specific geometries of each embodiment vary, all the embodiments accomplish the placement through their specific designs. As such, any combination of geometric entities used to create a stress relief cutout that accomplishes this placement is contemplated herein. Specifically, but not limiting, it is contemplated that the linear sections may be modified into spline curves, radii, multiple radii of differing magnitudes, or any combination thereof. Likewise, it is contemplated that and of the radii may be modified into spline curves, multiple radii of differing magnitudes, combinations of radii and straight or splined segments, or any combination thereof.

[0408] Another embodiment of a stress relief cutout **75** is shown in FIGS. **185** and **186**. Stress relief cutout **75** is formed on the hardened insert **1123**. In this embodiment the stress relief cutout **75** comprises a linear section **551**. The linear section **551** is tangent to the second radius **538** of the first bevel **426** and creates an angle β_6 with the front surface **416** as shown in FIG. **186**. In this

embodiment, the angle $\beta 6$ is 5 degrees but may be other non-zero values. Depending on the other values chosen, other modifications to the first bevel **426** may be required. The stress relief cutout **75** does not contact the annular shoulder **400**. In this embodiment the fluid routing plug **132** has only a transition radius **152** between the outer intermediate surface **336** and the annular shoulder **400**. The fluid routing plug **132** does not have a stress relief cutout.

[0409] The hardened insert **1123** contacts the annular shoulder **400** at the point where the linear section **551** joins the second radius **538**, as shown in FIG. **185**. This point is defined as the minimum radial contact point **P6**. The minimum radial contact point **P6** is at a radial distance **D6** from the section of the outer intermediate surface **336** of the fluid routing plug **132** that contains the second annular groove **392**. The second annular groove **392** is shown in FIG. **165**. The minimum radial contact point **P6** represents a circular line of contact that extends around the front surface **416** of hardened insert **1123**.

[0410] In operation, the radial distance **D6** locates the minimum radial contact point **P6** on the front surface **416** of the hardened insert **1123**, positioned radially outward from the inner intermediate surface **420**, toward the outer intermediate surface **422**. This reduces or eliminates chipping or spalling of the hardened insert **1123**.

[0411] Stress relief cutout **75** may be applied to hardened insert **210** if a non-tapered embodiment is used. While this embodiment of stress relief cutout **75** comprises only one component, the linear section **551**, other embodiments comprising multiple sections having geometric constructions similar to the embodiments **175**, **275**, **375**, **475**, and **575** may be used. As with the embodiments employed on the fluid routing plug **132** the key feature of this improvement is to locate the minimum radial contact point **P6** as far as practical from the inner intermediate surface **420** of the hardened insert **1123** or **210**.

[0412] The fluid end sections described herein have various embodiments of housings, inner components, and components attached to the various housings. While not specifically shown in a figure herein, various features from one fluid end section embodiment may be included in another fluid end section embodiment.

[0413] In alternative embodiments, features of one of the various housings, inner components, or components attached to the various housings and described in the '552 application, may be included in one or more of the fluid end sections disclosed herein. One of skill in the art will appreciate that the various housing and components described herein may have different shapes and sizes, depending on the shape and size of the various components chosen to assemble each fluid end section.

[0414] One or more kits may be useful in assembling a fluid end assembly out of the various fluid end sections described herein. A single kit may comprise a plurality of one of the various embodiments of housings and fasteners described herein. The kit may further comprise a plurality of one or more of the various inner components described herein. The kit may even further comprise a plurality of one or more of the various components attached to the various housings described herein.

[0415] The various features and alternative details of construction of the apparatuses described herein for the practice of the present technology will readily occur to the skilled artisan in view of the foregoing discussion. It is to be understood that even though numerous characteristics and advantages of various embodiments of the present technology have been set forth in the foregoing description, together with details of the structure and function of various embodiments of the technology, this detailed description is illustrative only. Changes may be made in detail, especially in matters of structure and arrangements of parts within the principles of the present technology to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

Claims

1. An apparatus, comprising: a fluid routing plug, comprising: an outer surface; an annular shoulder; and a stress relief cutout defined on a surface of the fluid routing plug between the outer surface and the annular shoulder; in which the annular shoulder is configured to abut an insert, the insert comprising a beveled surface.
2. A fluid end, comprising: the apparatus of claim 1; an insert which is the insert of claim 1; and a housing containing both the apparatus of claim 1 and the insert.
3. The fluid end of claim 2, in which the stress relief cutout is in a spaced relationship with the insert.
4. The fluid end of claim 2, in which the stress relief cutout does not engage the insert.
5. The fluid end of claim 2, in which the insert is made of a harder material than a material used to make the housing.
6. The fluid end of claim 2, in which the insert is made of tungsten carbide.
7. The fluid end of claim 6, in which the housing is made of stainless steel.
8. A fluid routing plug configured to be situated within a fluid end, the fluid routing plug comprising: a body having a first surface and an opposed second surface joined by an outer intermediate surface, the outer intermediate surface comprising: an annular shoulder formed therein; an annular groove formed therein; and a stress relief cutout formed therein and situated intermediate the annular shoulder and the annular groove; a longitudinal axis extending through the body and both the first and second surfaces; a plurality of first fluid passages interconnecting the first surface and the outer intermediate surface; and a plurality of second fluid passages interconnecting the first surface and the second surface.
9. The fluid routing plug of claim 8, in which the annular groove is configured to receive an annular seal.
10. The fluid routing plug of claim 8, in which the annular shoulder is configured to abut a wear ring; in which the stress relief cutout is in a spaced relationship with the wear ring.
11. The fluid routing plug of claim 8, in which the annular groove is defined as a first annular groove; in which the body further comprises a second annular groove formed on the intermediate surface, the second annular groove configured to receive an annular seal.
12. The fluid routing plug of claim 8, in which the stress relief cutout comprises: a planar section; and a curved section; in which the curved section is situated intermediate the outer intermediate surface and the planar section.
13. The fluid routing plug of claim 12, in which the planar section is situated intermediate the annular shoulder and the curved section.
14. A fluid end, comprising: a fluid end housing; the fluid routing plug of claim 8; and a wear ring abutting the annular shoulder of the body.
15. The fluid end of claim 14, in which the wear ring does not contact the stress relief cutout.
16. The fluid end of claim 14, in which the wear ring comprises: a first surface engaging the annular shoulder; a second surface opposed to the first surface and engaging a portion of the fluid end housing; an inner intermediate surface bound between the first and second surfaces; an outer intermediate surface bound between the first and second surfaces; and a first bevel formed intermediate the first surface and the outer intermediate surface.
17. The fluid end of claim 16, in which the wear ring further comprises: a second bevel situated intermediate the inner intermediate surface and the second surface.
18. The fluid end of claim 17, in which the first bevel is larger than the second bevel.
19. The fluid end of claim 14, in which the fluid end housing comprises: a longitudinal bore formed therein; and a valve pair situated within the longitudinal bore, the valve pair configured to engage the fluid routing plug.

- 20.** A fluid routing plug configured for use within a fluid end assembly comprising a housing and a wear ring, the fluid routing plug comprising: an annular seal engaging the wear ring; and a body defining an annular shoulder, the annular shoulder comprising: a cutout section; in which the cutout section does not contact the annular seal or the wear ring.
- 21.** An apparatus, comprising: a fluid end comprising a housing; a fluid routing plug situated within the housing; and a wear ring situated within the housing and configured to abut the fluid routing plug, the wear ring comprising: a first surface configured to abut the fluid routing plug; a second surface opposed to the first surface; an intermediate surface situated between the first and second surfaces; and a stress relief cutout situated between the intermediate surface and the first surface, in which the stress relief cutout does not contact the fluid routing plug.
- 22.** The apparatus of claim 21, in which the wear ring further comprises: a bevel situated between the intermediate surface and the stress relief cutout, in which the bevel does not contact the fluid routing plug.
- 23.** The apparatus of claim 22, in which the bevel comprises: a first radius joining the stress relief cutout; a second radius joining the intermediate surface; and a linear section situated between the first and second radii.
- 24.** The apparatus of claim 21, in which the stress relief cutout comprises: a planar section; in which the planar section is angled with respect to the first surface at a non-zero angle.
- 25.** The apparatus of claim 24, in which the non-zero angle is 5 degrees.
-